This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

nature portfolio

Peer Review File

Three-dimensional random-access confocal microscopy with 3D remote focusing system

Corresponding Author: Professor Shangguo Hou

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Li et al. demonstrates three-dimensional (3D) random-access confocal fluorescence imaging using a commercially available 3D dynamic focusing system (FL8220-3D-500, Sino-Galvo). Compared to a conventional 3D piezoelectric stage, the device enables high-speed imaging up with a refresh rate of up to 500 Hz for imaging three distant fluorescent beads, which is 25 times faster than a piezostage-based scanning method. Several specific aspects in the manuscript could be addressed to potentially clarify the significance of the work.

Technical aspects:

[1] In section “Configuration and 3D Resolution characterization of 3D-DyFI,” the optical principle of remote focusing is unclear. As a reader in optics field, I’d like to understand the details of the configuration of the plano-convex lens and two lenses in the remote focusing system. This includes: (1) the focal lengths and distances of each lens, and (2) specifications of the used galvo mirrors and linear actuators. The essential principles of remote focusing should be more elaborated in the main text, and the details about the dynamic focusing system should be clarified at least in the supplementary section.

[2] The reported FWHM shows poorer resolution in 3D-DyFI, but this is not clearly indicated in the manuscript. I strongly suspect that the intrinsic spherical aberrations due to the array of plano-concave lens and two convex lenses caused the resolution degradation. Such indications and attributions of poorer 3D-DyFI should be fairly mentioned in the revised manuscript.

[3] In Fig. 2, I am also interested in the bidirectional repeatability of the system. If possible, could you provide data on whether the galvo mirrors and the z-axis linear actuators exhibited any mechanical hysteresis? If so, how much bidirectional repeatability is guaranteed?

Contents aspects:

[4] Would you kindly consider merging the contents of Fig. 6 and 7? I believe that the authors can clearly compare the performance gap between 3D-DyFI and the conventional piezostage-based scanning method in a cleverer way. As a reader, I had difficulty comparing the performance between them in the current version of the manuscript.

In the introduction and discussion, some relevant and essential references are missing:

[5] For instance, the manuscript should explicitly clarify that the concept of the remote focusing is not completely new. For this, please add the following references: (i) Opt. Express 16, 21843-21848 (2008) (ii) Proceedings of the National Academy of Sciences 108.49 (2011): 19504-19509, (iii) Optics Express 29.5 (2021): 6293-6301. Please then compare how the proposed method in the manuscript is different from conventional methods and what advantages it has.

[6] Additionally, recent technical advances allow for commercially available spatial light modulators and tunable acoustic index gradient lens, which can also offer high-speed lateral random-access imaging and axial scanning, respectively. Please cite the following references: (i) eLife 11:e73266 (2022), (2) Optics letters 33.18 (2008): 2146-2148. What is the advantage of the proposed method compared to using such devices?

Reviewer #2

(Remarks to the Author)
In this manuscript, the authors propose a 3D random-access confocal microscopy with remote-focusing system, namely 3D-DyFI, as a technique able to perform 3D raster scanning imaging and 3D random-access imaging. To validate the performance of the proposed 3D-DyFI, the authors perform 3D imaging of fluorescent beads and Hela cells by considering the 3D piezo stage-based confocal microscopy as ground truth. In this way, they demonstrate that their solution is simple, cost-effective, and fast with respect to a standard technology. In particular, the authors demonstrate that their system has an isotropic response time across the x, y, and z axes, with a 34-fold improvement in axial response time over traditional piezo stages, and with the possibility to continuously monitoring the signals in three different planes, showing a refresh rate of 500 Hz on 12 two different positions in 3D.

The topic of this manuscript could be of interest for the readers of the Communications Engineering journal. In general, the manuscript is well written and easy to read and understand even by a reader who is not expert in this research topic. The authors present a complete and precise description of the methods, with all details needed to reproduce them. Most of the claims are well supported by the exhaustive validation proposed in the Results section. Therefore, in my opinion, this manuscript has the potential to be accepted in this journal, but the following major revisions must first be addressed.

1. In the abstract, the authors say that “this method is also compatible with multiphoton imaging and multicolor imaging in principle.” However, in the manuscript they don’t demonstrate this. Thus, this claim must be removed from the abstract and must be reported only in the Discussion section.
2. In the first sub-section of the Results, the authors measure the resolution of their system. It results that the resolution of the proposed 3D-DyFI system (344 nm × 518 nm × 1.02 μm) is worse than the piezoelectric stage (273 nm × 451 nm × 0.9 μm). However, throughout the manuscript, the authors only claim that the 3D-DyFI system is simple, cost-effective, and fast with respect to the piezoelectric stage. In my opinion, to be clear to the readers, the authors must add this aspect about resolution when they discuss the advantages of their system.
3. In the first sub-section of the Results, the authors say that “It is worth to note that due to the relatively large size of the fluorescent beads (200 nm), the measured FWHM value could be larger than the actual resolution of 3D-DyFI.” This means that the reported resolution (344 nm × 518 nm × 1.02 μm) is not the real one. In my opinion, it is needed to measure and report the real resolution of the system.
4. The resolution of the 3D-DyFI system is not isotropic in the three directions of the 3D space (344 nm × 518 nm × 1.02 μm), as well as the resolution of the piezoelectric stage (273 nm × 451 nm × 0.9 μm). Do the authors have any idea to make this resolution more isotropic? This could be an additional advantage compared to the piezoelectric stage.

Reviewer #3

(Remarks to the Author)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have addressed the important concerns raised in the original manuscript and significantly improved the manuscript. I recommend its publication for Communications Engineering.
One minor comment: Please revise the legend in Fig. 6 to clarify how different (a) and (b) are.

Reviewer #2

(Remarks to the Author)
The authors addressed my questions and I think that the manuscript is ready for publication in the current form.

Reviewer #3

(Remarks to the Author)
Reviewer 3 :

The authors have addressed my queries adequately. I recommend the paper for publication pending the following small query.

1. Figure 2 in the main manuscript may be either merged with figure 1 or shifted to Supplementary materials.

2. In the figure on page #28 of the rebuttal letter (which is Fig. 2C in the main manuscript), the authors show the plots for which the horizontal axis is in Volt. Is this true. Should it be in milli-Volts. Please check. This gives an erroneous minimum sampling of 0.29 nm i.e, 2.9 Angstrom.

Also note that the voltage output resolution of FPGA card is 0.3 mV. How the plots are shown in Volts.

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In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.
The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.
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We sincerely thank the editor and all reviewers for their valuable feedback that we have used to improve the quality of our manuscript. The reviewers' comments are laid out below in *italics* and specific concerns have been numbered. Our response is given in blue font and changes or additions to the manuscript are given in green font. The corresponding revision are highlighted in the revised manuscript.

Reviewer #1 (Remarks to the Author):

Li et al. demonstrates three-dimensional (3D) random-access confocal fluorescence imaging using a commercially available 3D dynamic focusing system (FL8220-3D-500, Sino-Galvo). Compared to a conventional 3D piezoelectric stage, the device enables high-speed imaging up with a refresh rate of up to 500 Hz for imaging three distant fluorescent beads, which is 25 times faster than a piezostage-based scanning method. Several specific aspects in the manuscript could be addressed to potentially clarify the significance of the work.

Response: Thank you for your comments and we greatly appreciate your professional review of our manuscript.

Technical aspects:

[1] In section "Configuration and 3D Resolution characterization of 3D-DyFI," the optical principle of remote focusing is unclear. As a reader in optics field, I'd like to understand the details of the configuration of the plano-convex lens and two lenses in the remote focusing system. This includes: (1) the focal lengths and distances of each lens, and (2) specifications of the used galvo mirrors and linear actuators. The essential principles of remote focusing should be more elaborated in the main text, and the details about the dynamic focusing system should be clarified at least in the supplementary section.

Response: Thank you for your valuable comments. We agree with you that the remote focusing system needs to be more detailed described. We now added a detailed schematic of the remote focusing system to the supplementary information. The focal length of each lens and their distances are described in the schematic. We also added the specifications of the used galvo mirrors and galvo motors in the supplementary information. It's worth to noted that the mirrors and lens are controlled by the same kind of galvo motor. In the introduction section, we added the essential principles of remote focusing imaging as your kind suggestion.

Revision 1: In page 2 of supplementary information, the schematic figure of remote focusing system is added:

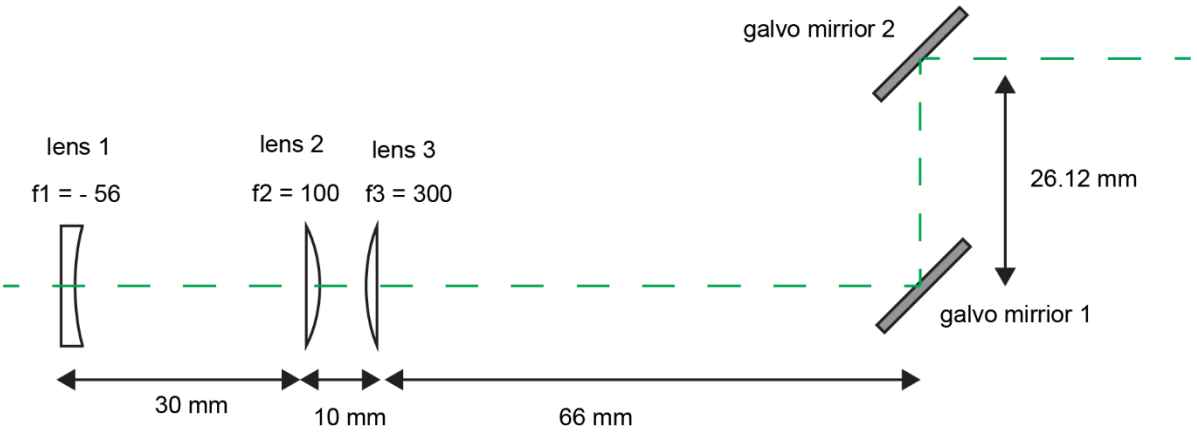


Figure S1. Detailed schematic of remote focusing system.

Revision 2: In page 15 of supplementary information, we added a **Supplementary Note 1** to describe the specifications of the galvo mirrors and motors in remote focusing system:

"**Supplementary Note 1. Specifications of the galvo mirrors and motors in remote focusing system.**

The mirrors are dielectric coated, with Ravg≥98% @ 405 - 700 nm at angle of incidence from 30 to 60 degrees. The size of the mirror is 30 mm × 35 mm. The linearity of motors is 99.9%. The long-term drift of motors over 8 hours: < 0.5 mRad. The machinery scan angle of XY scanning is ± 11°. The control voltage range is ± 5 V. The repeatability, indicated by relative root mean square error (RRMSE), are 0.058, 0.064, and 0.033 for x, y, and z, respectively (Figure S2 and S3)."

Revision 3: In Page 3 Line 11, the essential principles of remote focusing imaging is added:

"Remote focusing imaging, developed over a decade ago, is a technique that adjusts the focal plane of an optical system without requiring the mechanical movement of the sample[29-32]. This method employs auxiliary optical elements, such as remote objective with translational mirrors, to dynamically change the imaging focal plane with minimal aberration."

[2] The reported FWHM shows poorer resolution in 3D-DyFI, but this is not clearly indicated in the manuscript. I strongly suspect that the intrinsic spherical aberrations due to the array of plano-concave lens and two convex lenses caused the resolution degradation. Such indications and attributions of poorer 3D-DyFI should be fairly mentioned in the revised manuscript.

Response: Thank you for your valuable comments. We agree with you that the degraded resolution of 3D-DyFI may be caused by the spherical aberrations from one plano-concave lens and two convex lenses. In the future, we can replace these lenses with achromatic doublets lenses to reduce the spherical aberrations.

Revision: In Page 5 Line 6, the following sentences are added to clarify the resolution degradation:

"The resolution of 3D-DyFI shows slightly worse than piezoelectric stage scanning result, which could be attributed to the spherical aberrations from one plano-concave lens and two convex lenses in dynamic focusing system. This can be improved by replacing these lenses with achromatic doublets lenses in the future."

[3] In Fig. 2, I am also interested in the bidirectional repeatability of the system. If possible, could you provide data on whether the galvo mirrors and the z-axis linear actuators exhibited any mechanical hysteresis? If so, how much bidirectional repeatability is guaranteed?

Response: This is a good question. As your nice suggestion, we added the bidirectional repeatability measurement results in the supporting information file. We conducted the repeatability test in two ways. Firstly, we measured the bidirectional repeatability via power measurement after a pinhole (**Figure S2**). This method works very well for axial bidirectional repeatability measurement. However, it can only measure a small range in x and y direction. Therefore, we used the scanning reflected images to further characterize the bidirectional repeatability in x and y direction with large range (**Figure S3**). As you can see from the measurement results, we didn't find obvious mechanical hysteresis. The repeatability, indicated by relative root mean square error (RRMSE), are 0.058, 0.064, and 0.033 for x, y, and z, respectively.

Revision: In page 3 to page 5 of supplementary information, the repeatability measurement results are added:

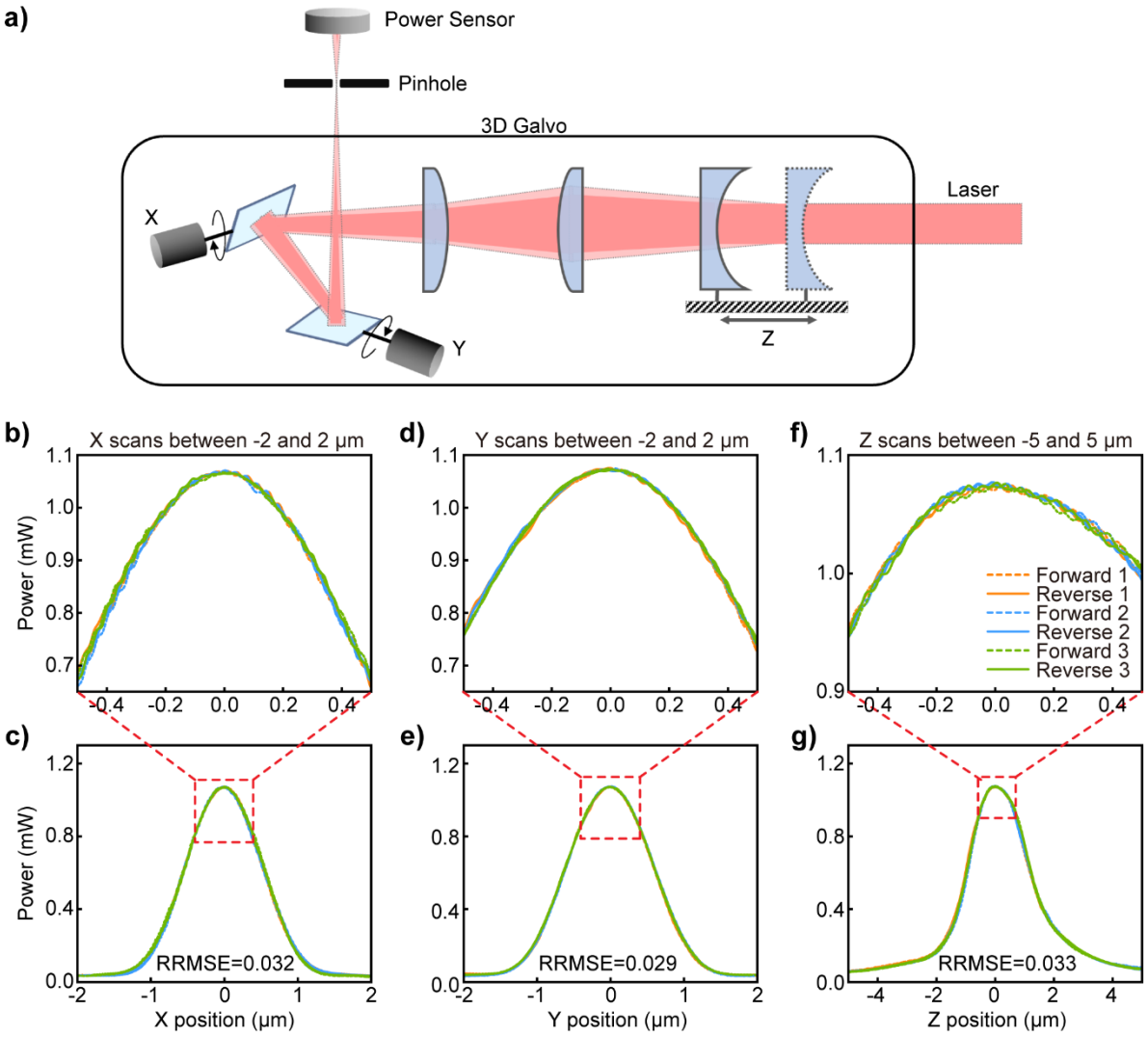


Figure S2. Bidirectional repeatability measurement of the 3D dynamic focusing system via power measurement after a pinhole. (a) Schematic of setup for testing the bidirectional repeatability of the 3D dynamic focusing system. A 100 μm pinhole was placed near the focal plane of 3D dynamic focusing system and a power meter was used to detect the light intensity after the pinhole. **(b-g)** The laser power measurement results in x **(b-c)**, y **(d-e)**, and z **(f-g)**, respectively. Three measurements were conducted, with forward scans represented by orange, blue, and green dashed lines, and reverse scans by solid lines. **(c, e, g)** Scanning results along the x, y, and z axes. The scanning range with ranges for x and y is -2 μm to 2 μm. The scanning range with ranges for z is -5 μm to 5 μm. (**b**, **d**, **f**) Enlarged views of the red dashed boxes in **(c, e,**

g), respectively. The relative root mean square error (RRMSE) values were calculated for characterizing the repeatbility. The RRMSE values for the x, y, and z axes were 0.032, 0.029, and 0.033, respectively, indicating good bidirectional repeatability across all axes.

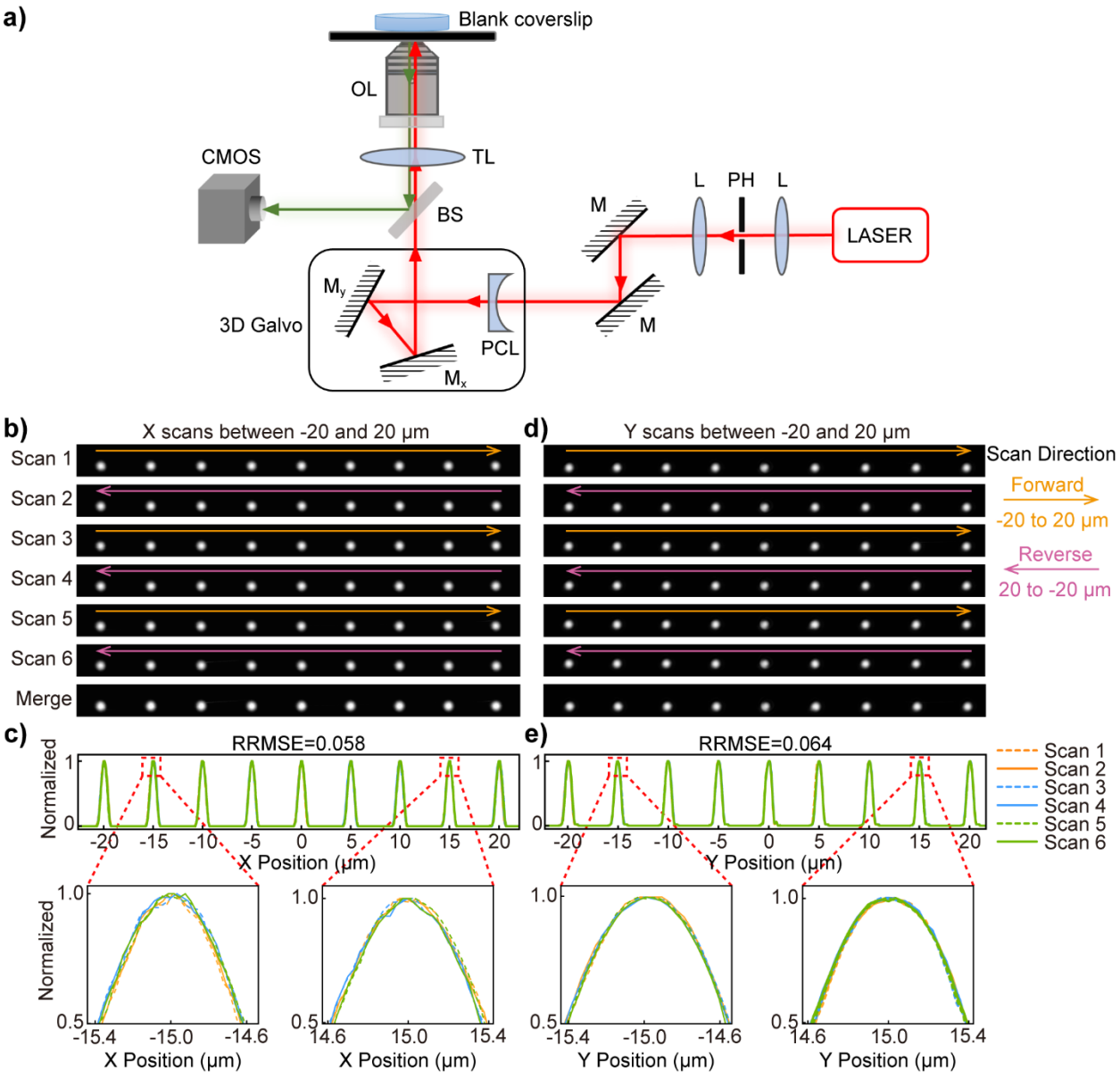


Figure S3. Bidirectional repeatability measurement of the 3D dynamic focusing system via scanning images. (a) Schematic of setup for testing the bidirectional repeatability of the 3D dynamic focusing system. The excitation light is focused on a blank coverslip and imaged on a CMOS camera. As the system scans in the x and y directions, the CMOS captures laser spot

images at 5 μm intervals. **(b, d)** Scanning images in the range between -20 and 20 μm in x and y direction, respectively. Three sets of scanning results are shown: Scan 1 and Scan 2 represent the first set, Scan 3 and Scan 4 the second set, and Scan 5 and Scan 6 the third set. Each set shows both forward (orange arrow) and reverse (pink arrow) scan results. The bottom row presents the merged images from all six scans. **(c, e)** Overlayed normalized intensity profiles along the scanning direction. The RRMSE values for the x and y axes are 0.058 and 0.064, respectively.

Contents aspects:

[4] Would you kindly consider merging the contents of Fig. 6 and 7? I believe that the authors can clearly compare the performance gap between 3D-DyFI and the conventional piezostage-based scanning method in a cleverer way. As a reader, I had difficulty comparing the performance between them in the current version of the manuscript.

Response: Thank you for your nice suggestion. We apologize for the inefficient figure arrangement in previous manuscript. Now we have merged **Fig. 6** and **Fig. 7** for better visualization. We removed the original Fig. 6c,e,g,h and Fig. 7c,e,g,h for clarity since they show only redundant information.

Revision: The Fig. 6 and Fig. 7 has been merged into a new Fig. 7:

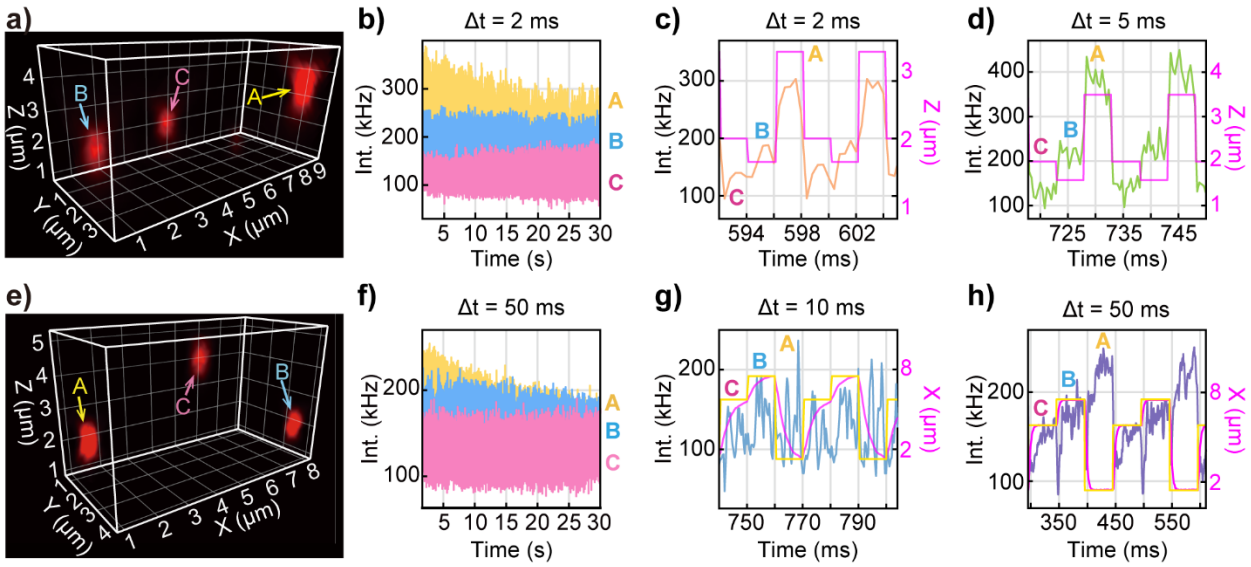


Fig. 7. 3D Random-access imaging with 3D-DyFI and piezo stage-based confocal system. (a) Three-dimensional image of three fluorescent beads embedded in agarose obtained with 3D-DyFI. The particles are labeled with A, B, and C, respectively. **(b)** The segmentary intensity trace of the three fluorescent beads in **(a)**. The illumination time in each period is 2 ms. The lines in different colors indicate different particles as labeled in **(a)**. **(c-d)** The continuous intensity trace from three fluorescent beads with 2 ms illumination time **(c)** and 5 ms illumination **(d)** in each imaging period. The magenta lines show the z position of laser focus, obtained from the input of 3D dynamics focusing system. The fluorescence intensity is bin in 400 μs, considered the ~ 800 μs response time of 3D dynamic focusing system. **(e)** Three-dimensional image of three fluorescent beads embedded in agarose obtained with piezo stage-based confocal system. The particles are labeled with A, B, and C, respectively. **(f)** The segmentary intensity trace of the three fluorescent beads in **(e)**. The illumination time in each period is 50 ms. The lines in different colors indicate different particles as labeled in **(e)**. **(g-h)** The continuous intensity trace from three fluorescent beads with 10 ms illumination time **(g)** and 50 ms illumination **(h)** in each imaging period. The magenta lines show the readout position from the piezoelectric stage controller. The yellow line indicates the input position of the piezoelectric stage controller.

In the introduction and discussion, some relevant and essential references are missing:

[5] For instance, the manuscript should explicitly clarify that the concept of the remote focusing is not completely new. For this, please add the following references: (i) Opt. Express 16, 21843-21848 (2008) (ii) Proceedings of the National Academy of Sciences 108.49 (2011): 19504-19509, (iii) Optics Express 29.5 (2021): 6293-6301. Please then compare how the proposed method in the manuscript is different from conventional methods and what advantages it has.

Response: Thank you for your comment. In the revised manuscript we have clarified that remote focusing imaging is a developed technique, and the suggested references have been added to the manuscript.

The remote focusing methods on the above references utilize a remote objective and a translational mirror to shift the focal plane. In these methods, a polarization beam splitter is also utilized in the detection path, which can bring 50% fluorescence loss. As contrast, 3D-DyFI doesn’t suffer from this problem and has higher photon collection efficiency.

Revision 1: In Page 3 Line 11, the following sentences has been added to clarify the remote focusing imaging is a developed technique and the relative references have been added:

“Remote focusing imaging, developed over a decade ago, is a technique that adjusts the focal plane of an optical system without requiring the mechanical movement of the sample[29-32]. This method employs auxiliary optical elements, such as remote objective with translational mirrors, to dynamically change the imaging focal plane with minimal aberration[29-31, 33].”

“29. Botcherby, E.J., Juškaitis, R., Booth, M.J. & Wilson, T. An optical technique for remote focusing in microscopy. Optics Communications 281, 880-887 (2008).

30. Botcherby, E.J., Juskaitis, R., Booth, M.J. & Wilson, T. Aberration-free optical refocusing in high numerical aperture microscopy. Opt. Lett. 32, 2007-2009 (2007).

31. Botcherby, E.J., Booth, M.J., Juškaitis, R. & Wilson, T. Real-time extended depth of field microscopy. Optics Express 16, 21843-21848 (2008).

32. Anselmi, F., Ventalon, C., Bègue, A., Ogden, D. & Emiliani, V. Three-dimensional imaging and photostimulation by remote-focusing and holographic light patterning. Proceedings of the National Academy of Sciences 108, 19504-19509 (2011).

33. Hugonnet, H., Lee, M. & Park, Y. Optimizing illumination in three-dimensional deconvolution microscopy for accurate refractive index tomography. Optics Express 29, 6293-6301 (2021).”

Revision 2: In Page 16 Line 30, the following sentences are added to compare the remote objective based-remote focusing methods and 3D-DyFI:

“In the remote objective based-remote focusing methods, a translational mirror and a polarization beam splitter are utilized in the detection path. The use of polarization beam splitter results in 50 % fluorescence loss in these methods. As contrast, 3D-DyFI doesn’t suffer from this problem and has higher photon collection efficiency.”

[6] Additionally, recent technical advances allow for commercially available spatial light modulators and tunable acoustic index gradient lens, which can also offer high-speed lateral random-access imaging and axial scanning, respectively. Please cite the following references: (i) eLife 11:e73266 (2022), (2) Optics letters 33.18 (2008): 2146-2148. What is the advantage of the proposed method compared to using such devices?

Response: Thank you for your nice suggestion. We added the discussion about the applications of SLM and TAG lens on 3D random-access imaging and fast axial scanning in the "Conclusion and Discussion" section. The suggested references have been added to the manuscript.

You are correct that the spatial light modulator (SLM) and digital micromirror device (DMD) have also been frequently used in random-access imaging. However, the degrees of freedom (DoF) in these systems inherently limit the number of accessible voxels, even with the notable improvements in the recently reported three-dimensional multi-site random access photostimulation (3D-MAP) method (eLife 2022, 11:e73266). In contrast, in 3D-DyFI, the laser focus position is determined by the input voltage to the 3D dynamic focusing system, which eliminates this limitation.

The TAG lens can oscillate the laser focus sinusoidally along the axial direction at frequencies reaching hundreds of kilohertz. However, the TAG lens is unable to maintain the laser focus at a specific focal plane for extended periods, and pulsed light is required for selective focal plane excitation. In 3D-DyFI, the focal position is controlled through analog voltage, allowing for more straightforward and precise manipulation of the excitation focus.

Revision: In Page 17 Line 16, the following discussion has been added:

The spatial light modulator (SLM) and digital micromirror device (DMD) have also been used to generate 3D scanning patterns[32, 38, 39]. However, the number of degrees of freedom (DoF) in these systems inherently limits the number of accessible voxels, despite significant improvements with the recently reported three-dimensional multi-site random access photostimulation (3D-MAP) method[38]. In contrast, in 3D-DyFI, the laser focus position is determined by the input voltage to the 3D dynamic focusing system, which eliminates this limitation.

Another commonly used optical device for rapid axial scanning is the tunable acoustic gradient index of refraction (TAG) lens[40-43]. The TAG lens can oscillate the laser focus sinusoidally along the axial direction at frequencies reaching hundreds of kilohertz. However, the TAG lens is unable to maintain the laser focus at a specific focal plane for extended periods, and pulsed light is required for selective focal plane excitation. In 3D-DyFI, the focal position is controlled through analog voltage, allowing for more straightforward and precise manipulation of the excitation focus.

38. Xue, Y., Waller, L., Adesnik, H. & Pégard, N. Three-dimensional multi-site random access photostimulation (3D-MAP). Elife 11, e73266 (2022).

39. Lutz, C. et al. Holographic photolysis of caged neurotransmitters. Nature Methods 5, 821-827 (2008).

40. Mermillod-Blondin, A., McLeod, E. & Arnold, C.B. High-speed varifocal imaging with a tunable acoustic gradient index of refraction lens. Opt. Lett. 33, 2146-2148 (2008).

41. Deguchi, T. et al. Volumetric Lissajous confocal microscopy with tunable spatiotemporal resolution. Biomed. Opt. Express 11, 6293-6310 (2020).

42. Kong, L. et al. Continuous volumetric imaging via an optical phase-locked ultrasound lens. Nature Methods 12, 759-762 (2015).

43. Duocastella, M., Vicidomini, G. & Diaspro, A. Simultaneous multiplane confocal microscopy using acoustic tunable lenses. Optics Express 22, 19293-19301 (2014).

Reviewer #2 (Remarks to the Author):

In this manuscript, the authors propose a 3D random-access confocal microscopy with remote-focusing system, namely 3D-DyFI, as a technique able to perform 3D raster scanning imaging and 3D random-access imaging. To validate the performance of the proposed 3D-DyFI, the authors perform 3D imaging of fluorescent beads and Hela cells by considering the 3D piezo stage-based confocal microscopy as ground truth. In this way, they demonstrate that their solution is simple, cost-effective, and fast with respect to a standard technology. In particular, the authors demonstrate that their system has an isotropic response time across the x, y, and z axes, with a 34-fold improvement in axial response time over traditional piezo stages, and with the possibility to continuously monitoring the signals in three different planes, showing a refresh rate of 500 Hz on 12 two different positions in 3D.

The topic of this manuscript could be of interest for the readers of the Communications Engineering journal. In general, the manuscript is well written and easy to read and understand even by a reader who is not expert in this research topic. The authors present a complete and precise description of the methods, with all details needed to reproduce them. Most of the claims are well supported by the exhaustive validation proposed in the Results section. Therefore, in my opinion, this manuscript has the potential to be accepted in this journal, but the following major revisions must first be addressed.

Response: Thank you for your comments and we greatly appreciate your professional review of our manuscript.

1. In the abstract, the authors say that "this method is also compatible with multiphoton imaging and multicolor imaging in principle." However, in the manuscript they don't demonstrate this. Thus, this claim must be removed from the abstract and must be reported only in the Discussion section.

Response: Thank you for your comment. We have removed this sentence in current version of manuscript.

Revision: In abstract section, the following sentence has been removed:

"Additionally, this method is also compatible with multiphoton imaging and multicolor imaging in principle."

2. In the first sub-section of the Results, the authors measure the resolution of their system. It results that the resolution of the proposed 3D-DyFI system (344 nm × 518 nm × 1.02 μm) is worse than the piezoelectric stage (273 nm × 451 nm × 0.9 μm). However, throughout the manuscript, the authors only claim that the 3D-DyFI system is simple, cost-effective, and fast with respect to the piezoelectric stage. In my opinion, to be clear to the readers, the authors must add this aspect about resolution when they discuss the advantages of their system.

Response: Thank you for your comment. We have revised the manuscript to clarify that the resolution of 3D-DyFI is worse than piezoelectric stage scanning result. We think the degraded resolution of 3D-DyFI may be caused by the spherical aberrations from one plano-concave lens and two convex lenses. In the future, we can replace these lenses with achromatic doublets lenses to reduce the spherical aberrations.

Revision: In Page 5 Line 6, the following sentences are added to clarify the resolution degradation:

"The resolution of 3D-DyFI shows slightly worse than piezoelectric stage scanning result, which could be attributed to the spherical aberrations from one plano-concave lens and two convex lenses in dynamic focusing system. This can be improved by replacing these lenses with achromatic doublets lenses in the future"

3. In the first sub-section of the Results, the authors say that "It is worth to note that due to the relatively large size of the fluorescent beads (200 nm), the measured FWHM value could be larger than the actual resolution of 3D-DyFI." This means that the reported resolution (344 nm × 518 nm × 1.02 μm) is not the real one. In my opinion, it is needed to measure and report the real resolution of the system.

Response: Thank you for your comment. This is a good suggestion. We now have measured the PSF with 50 nm fluorescent bead. The size of 50 nm fluorescent bead is much smaller than the resolution of 3D-DyFI. Therefore, it can be effectively used to characterize the resolution of 3D-

DyFI. The measured results reveal a resolution of 301 nm × 330 nm × 0.96 μm for 3D-DyFI and 311 nm × 322 nm × 0.93 μm for piezoelectric stage scanning in x, y and z. We have replaced the PSF figures with measuring results with 50 nm fluorescent bead in **Figure 1**.

Revision 1: The Figure 1 have been renewed:

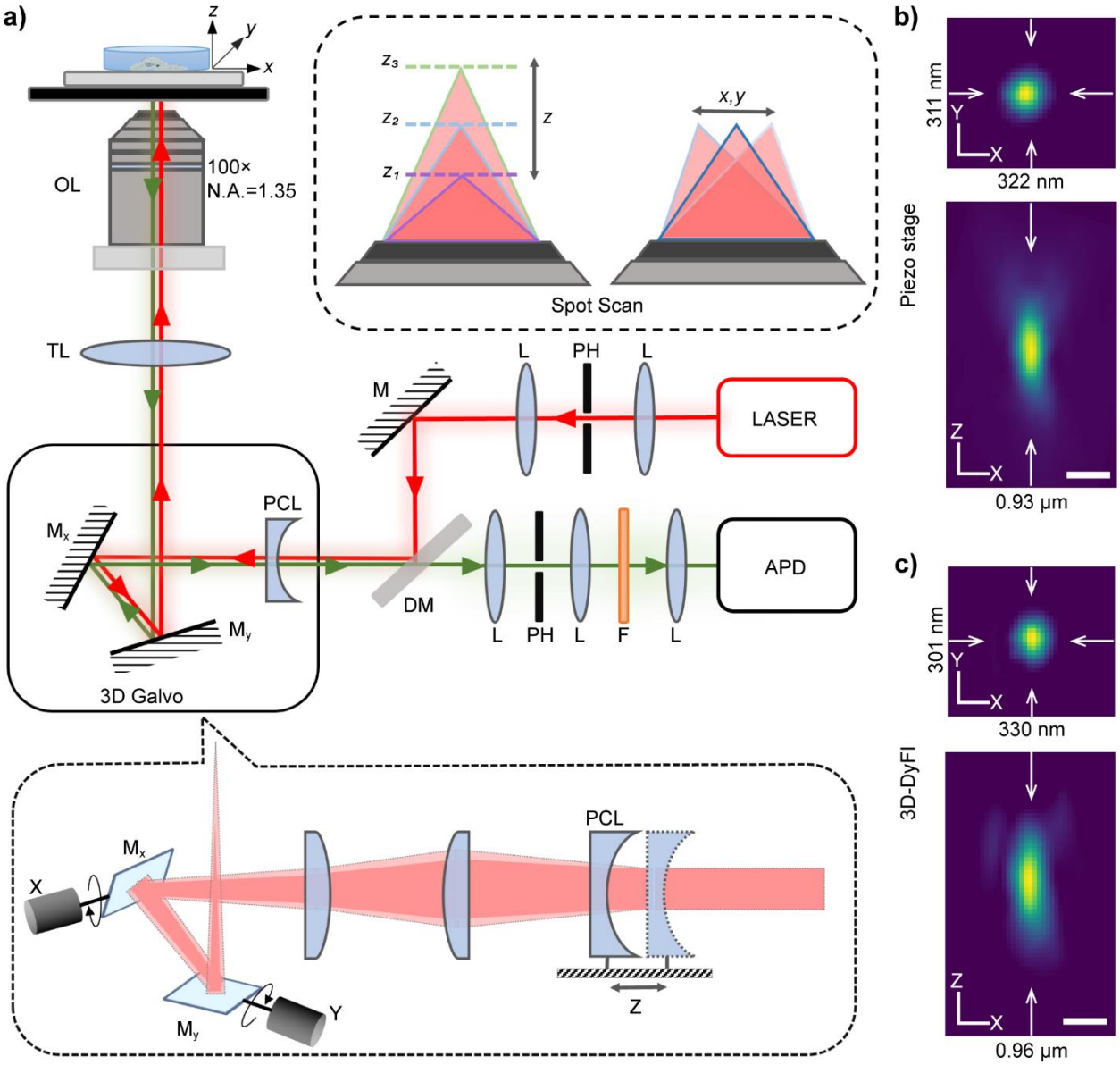


Fig. 1. Schematic and point spread function of the 3D-DyFI. **(a)** Schematic of 3D-DyFI. OL: Objective Lens, M: Mirror, F: fluorescence filter, DM: Dichroic Mirror, PH: Pinhole, PCL: Plano-Concave Lens, APD: Avalanche Photodiode. **(b-c)** The x-y scanning images (up panel)

and x-z scanning images (low panel) of 50 nm green fluorescent microsphere with **(b)** piezoelectric stage scanning and **(c)** 3D-DyFI. Scale bar: 500 nm.

Revision 2: In page 5 Line 3, the following sentence has been revised:

"The full widths at half maximum (FWHM) of intensity profiles of the fluorescent microspheres were subsequently measured (**Supplementary Figure S4**), revealing resolutions of 301 nm × 330 nm × 0.96 μm for 3D-DyFI and 311 nm × 322 nm × 0.93 μm for piezoelectric stage scanning in x, y and z, respectively."

Revision 3: The following sentence has been removed:

"It is worth to note that due to the relatively large size of the fluorescent beads (200 nm), the measured FWHM value could be larger than the actual resolution of 3D-DyFI".

4. The resolution of the 3D-DyFI system is not isotropic in the three directions of the 3D space (344 nm × 518 nm × 1.02 μm), as well as the resolution of the piezoelectric stage (273 nm × 451 nm × 0.9 μm). Do the authors have any idea to make this resolution more isotropic? This could be an additional advantage compared to the piezoelectric stage.

Response: Thank you for your comment. It is an excellent question. There are several ways to achieve isotropic resolution imaging. Some methods that most related to 3D-DyFI are listed as below:

(a) By placing a mirror after the focal point of objective and shaping the incident beam with spatial light modulator (SLM) or phase plates, the isotropic focusing can be achieved (*Physical Review Letters* 105, 203903, (2010); *Opt. Express* 21, 19701-19708 (2013); Appl. Opt. 53, 7838-7844 (2014*); Nat Biotechnol* 41, 1307–1319 (2023).). However, this kind of method does not fit 3D-DyFI, which requires laser focus scanning without moving sample.

(b) The isotropic resolution imaging can also be achieved via dual-view plane illumination method with switching illumination and detection between two perpendicular objectives

(*Nat Biotechnol* 31, 1032–1038 (2013)). This method requires computationally fusing the two perpendicular volumetric views. As the sample is stationary during imaging process as in 3D-DyFI, we think the 3D-DyFI can achieve isotropic resolution imaging in this way by add a perpendicular imaging path.

(c) The recently developed deep-learning methods can also improve the axial resolution (*Light Sci Appl* **12**, 204 (2023); Biomed. *Opt. Express* 11, 7273-7285 (2020); *Nat Commun* **13**, 3297 (2022); *Nat Commun* 12, 107 (2021);). This kind of methods doesn't require the optical modification and is applicable to 3D-DyFI.

Revision: The following sentences are added to the Conclusion and Discussion section:

Furthermore, as the sample is stationary during imaging process as in 3D-DyFI, it has the potential to achieve isotropic resolution imaging via dual-view plane illumination method with switching illumination and detection between two perpendicular objectives[52]. The recently developed deep-learning methods can also be adopted to improve the axial resolution[53-56].

52. Wu, Y. et al. Spatially isotropic four-dimensional imaging with dual-view plane illumination microscopy. Nature Biotechnology 31, 1032-1038 (2013).

53. Ning, K. et al. Deep self-learning enables fast, high-fidelity isotropic resolution restoration for volumetric fluorescence microscopy. Light: Science & Applications 12, 204 (2023).

54. Zhao, F. et al. Deep-learning super-resolution light-sheet add-on microscopy (Deep-SLAM) for easy isotropic volumetric imaging of large biological specimens. Biomed. Opt. Express 11, 7273-7285 (2020).

55. Park, H. et al. Deep learning enables reference-free isotropic super-resolution for volumetric fluorescence microscopy. Nature Communications 13, 3297 (2022).

56. Fang, C. et al. Minutes-timescale 3D isotropic imaging of entire organs at subcellular resolution by content-aware compressed-sensing light-sheet microscopy. Nature Communications 12, 107 (2021).

Reviewer #3

Review Report:

The manuscript titled, "three dimensional random-access confocal microscopy with 3D remote focusing system" by Li et al discuss a rapid 3D random access remote focusing technique for volume imaging. The technique claims multifold improvement in response time along x, y, z axes. Moreover, the technique achieves scanning without physically moving the sample and without mechanical movement. This is good for a range of applications. While the concept is good and economical, the application part needs to improve for the advantages of the technique to be apparent. Here are my major and minor comments:

Response: We sincerely appreciate your valuable comments and thorough review of our manuscript. Your expertise and insights have greatly contributed to the improvement of our work.

Major:

1. It would be nice to see the depth imaging capability of the proposed system on a cellular system.

Response: Thank you for your nice suggestion. For now, we only test the system with cell samples. We have added several cellular 3D imaging results to the manuscript and supplementary information, including mitochondria (Fig. 6), cell microtubule (Supplementary Fig. S8), cell membrane (Supplementary Fig. S9) and lipid droplets in live cell (Supplementary Fig. S10). Theoretically, the depth imaging capability of 3D-DyFI should be similar to conventional confocal microscopy, which is limited to the wavelength of excitation laser and the objective lens. It worth to mention in here that the axial imaging range of 3D-DyFI is limited to 10 μm.

Here we performed the axial slicing imaging of part of cell membrane in thick cell. The cells were initially imaging with piezo stage-based confocal microscopy to confirm the thickness of cell. Subsequently, the 3D-DyFI was used to image the top part of the cell to demonstrate its depth imaging capability.

Revision: The **Supplementary Figure S11** has been added to the supplementary information to demonstrate the imaging capability of 3D-DyFI in thick cells:

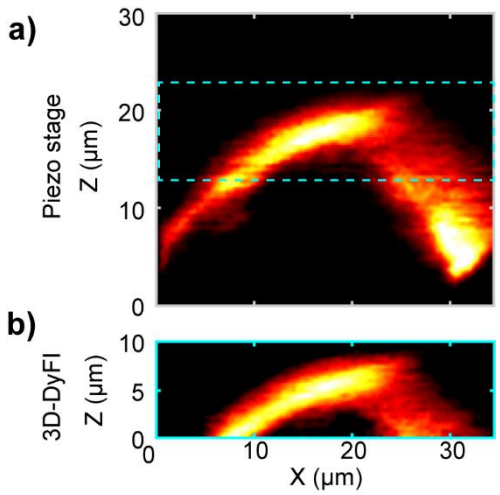


Figure S11. Thick cell imaging with 3D-DyFI. **(a)** x-z plane image of DiD-stained Hela cell membrane obtained with piezo stage-based confocal microscopy. (b) x-z plane image of DiD-stained Hela cell membrane obtained with 3D-DyFI. The imaging area is selected to the top part of cell as marked with dash box. Pixel size in x: 0.1 μm; Pixel size in z: 0.2 μm;

2. The schematic diagram needs more discussion with special attention to the 3D galvo section. Specifically discuss how PCL on linear actuator and Galvo mirror operated synchronously during data acquisition.

Response: We greatly appreciate your nice suggestion. We agree with you that remote focusing system needs to be more detailed described. We now added the detailed information about remote focusing system to the supplementary information (**Supplementary Figure S1-S3**). The focal length of each lens and their distances are described in the **Supplementary Figure S1**. we also added a **Supplementary Note 1** to describe the specifications of the galvo mirrors and motors in remote focusing system.

In the **Methods** section, we added the details of the remote focusing system operation and signal acquisition with data acquisition card. The motors for driving PCL and the galvo mirrors are independently controlled via three analog voltage output channels of the data acquisition card. Custom- developed software developed in LabVIEW is employed to set the analog voltages applied to the three axes of the remote focusing system. Specifically, the

voltages applied to the motors for PCL and galvo mirrors determine the position of laser focus position after objective lens. In random-access imaging mode, the remote focusing system sequentially switches laser focus among several predefined positions. In random-access imaging mode, the remote focusing system moves the laser focus in a raster scanning pattern. At the same time, the fluorescence intensity signal from avalanche photodiode (APD) detector is recorded on the digital input channel of the data acquisition card. Both the applied analog voltages to the remote focusing system and the fluorescence intensity signals are logged by the data acquisition card with precise timing tags.

Revision 1: In page 2 of supplementary information, the schematic figure of remote focusing system is added:

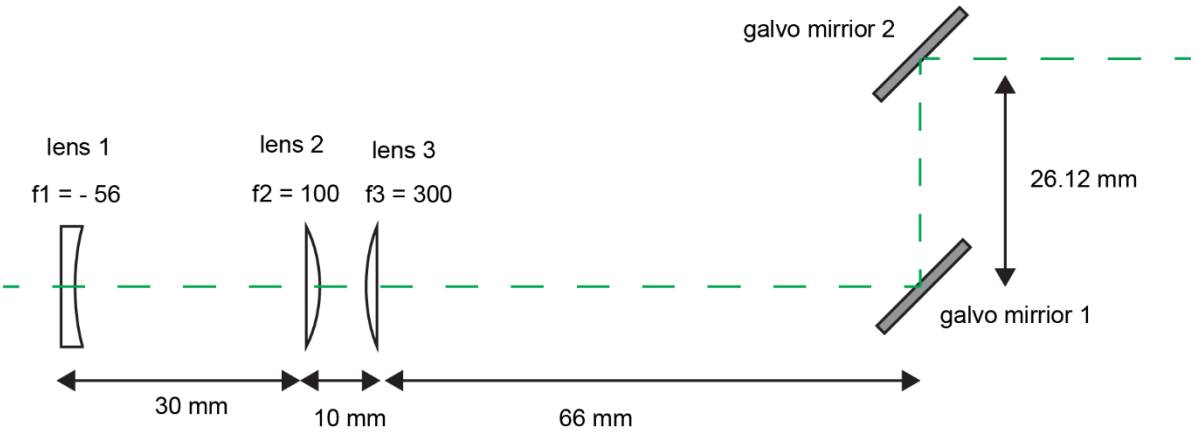


Figure S1. Detailed schematic of remote focusing system.

Revision 2: In page 15 of supplementary information, we added a Supplementary Note to describe the specifications of the galvo mirrors and motors in remote focusing system:

"**Supplementary Note 1. Specifications of the galvo mirrors and motors in remote focusing system.**

The mirrors are dielectric coated, with Ravg≥98% @ 405 - 700 nm at angle of incidence from 30 to 60 degrees. The size of the mirror is 30 mm × 35 mm. The linearity of motors is 99.9%. The long-term drift of motors over 8 hours: < 0.5 mRad. The machinery scan angle of XY scanning is ± 11°. The control voltage range is ± 5 V. The repeatability, indicated by relative

root mean square error (RRMSE), are 0.058, 0.064, and 0.033 for x, y, and z, respectively (Figure S2 and S3).”

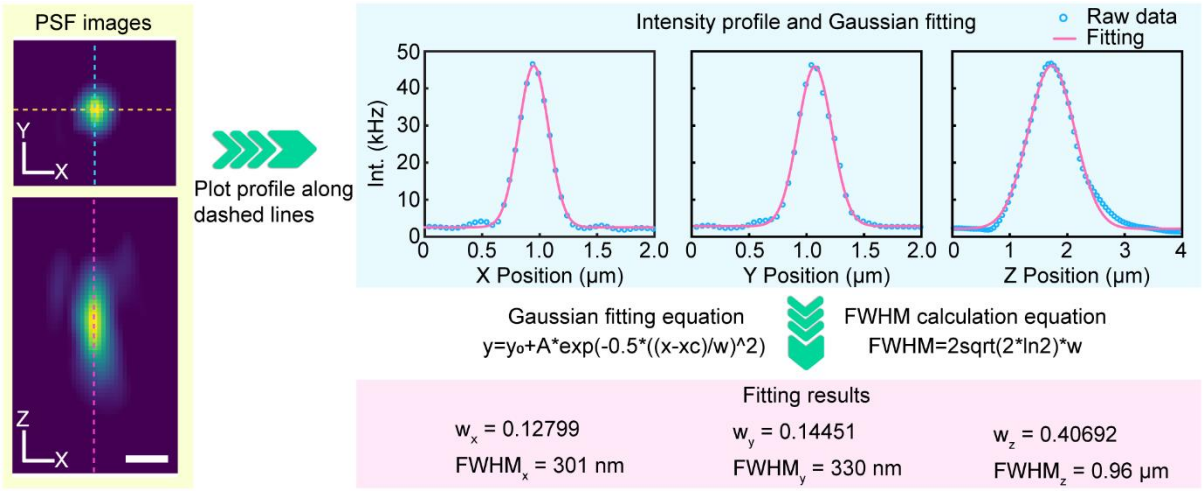
Revision 3: In page 24 Line 6, the following paragraph is added:

“***Remote focusing system operation***

The motors for driving PCL and the galvo mirrors are independently controlled via three analog voltage output channels of the data acquisition card. Custom-developed software developed in LabVIEW is employed to set the analog voltages applied to the three axes of the remote focusing system. Specifically, the voltages applied to the motors for PCL and galvo mirrors determine the position of laser focus position after objective lens. In random-access imaging mode, the remote focusing system sequentially switches laser focus among several predefined positions. In random-access imaging mode, the remote focusing system moves the laser focus in a raster scanning pattern. At the same time, the fluorescence intensity signal from avalanche photodiode (APD) detector is recorded on the digital input channel of the data acquisition card. Both the applied analog voltages to the remote focusing system and the fluorescence intensity signals are logged by the data acquisition card with precise timing tags.”

3. Needs more detail on the resolution of the proposed system with additional data.

Response: Thank you for your comment. We measured the resolution by fitting the full widths at half maximum (FWHM) of intensity profiles of fluorescent beads. The detailed fitting process is shown in the figure below:



Additionally, we have added more resolution characterization data by measuring the point spread function (PSF) with 50 nm green and red fluorescent beads. Due to their smaller size, these 50 nm beads are expected to provide more precise resolution characterization compared to 200 nm beads. The resolution was evaluated using two types of 50 nm beads: green fluorescent beads excited by a 488 nm laser and red fluorescent beads excited by a 635 nm laser. We have updated the images in Figures 1(**b-c**) and **Figure 3** to include the results obtained with 50 nm fluorescent beads. The resolution characterization results with 50 nm red fluorescent beads have been added to **Supplementary Figure S6-S7**.

Moreover, a comparison of the PSF profiles between 3D-DyFI and piezo-stage based confocal microscopy has been incorporated into the Supplementary Information.

Revision 1: The resolution characterization method has been added to the Supplementary Information:

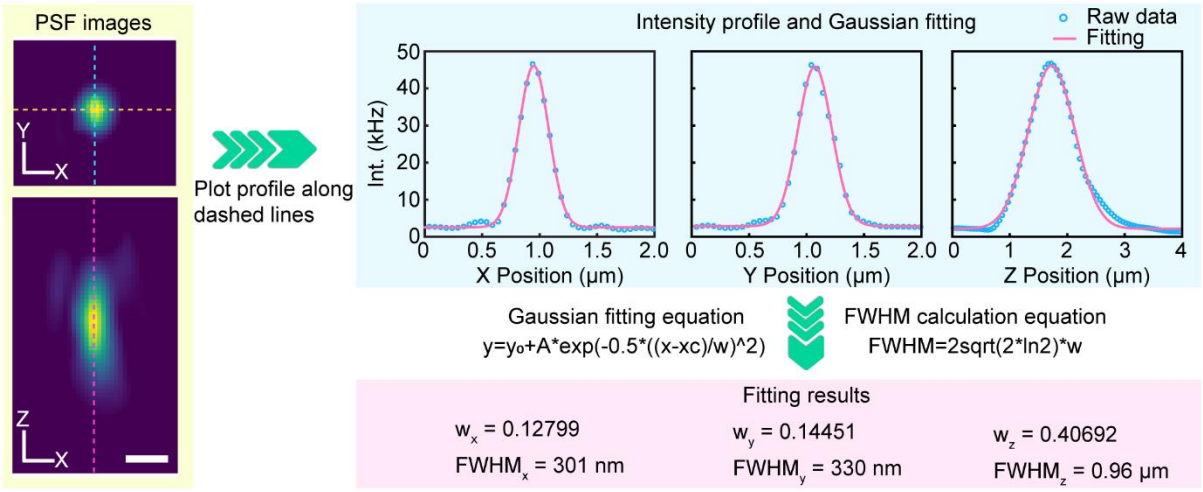


Figure S4. Resolution characterization with fluorescent sphere image. The intensity profiles along the x, y, and z directions are extracted and Gaussian fitted. The resolution is calculated by the full widths at half maximum (FWHM) as noted in the figure.

Revision 2: **Figure 1** and **Figure 3** have been replaced with the following figures:

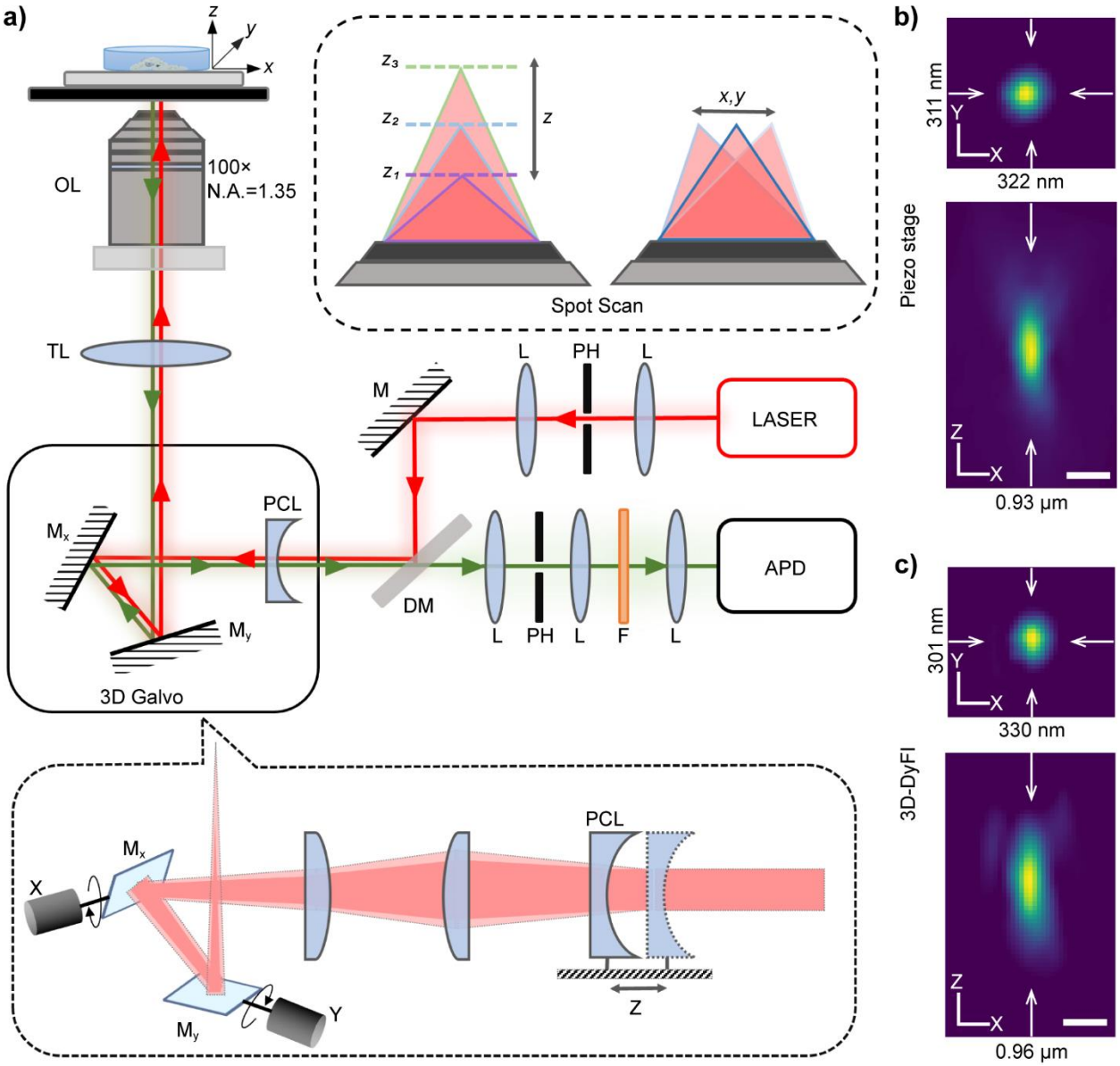


Fig. 1. Schematic and point spread function of the 3D-DyFI. (a) Schematic of 3D-DyFI. OL: Objective Lens, M: Mirror, F: fluorescence filter, DM: Dichroic Mirror, PH: Pinhole, PCL: Plano-Concave Lens, APD: Avalanche Photodiode. **(b-c)** The x-y scanning images (up panel) and x-z scanning images (low panel) of 50 nm green fluorescent microsphere with **(b)** piezoelectric stage scanning and **(c)** 3D-DyFI. Scale bar: 500 nm.

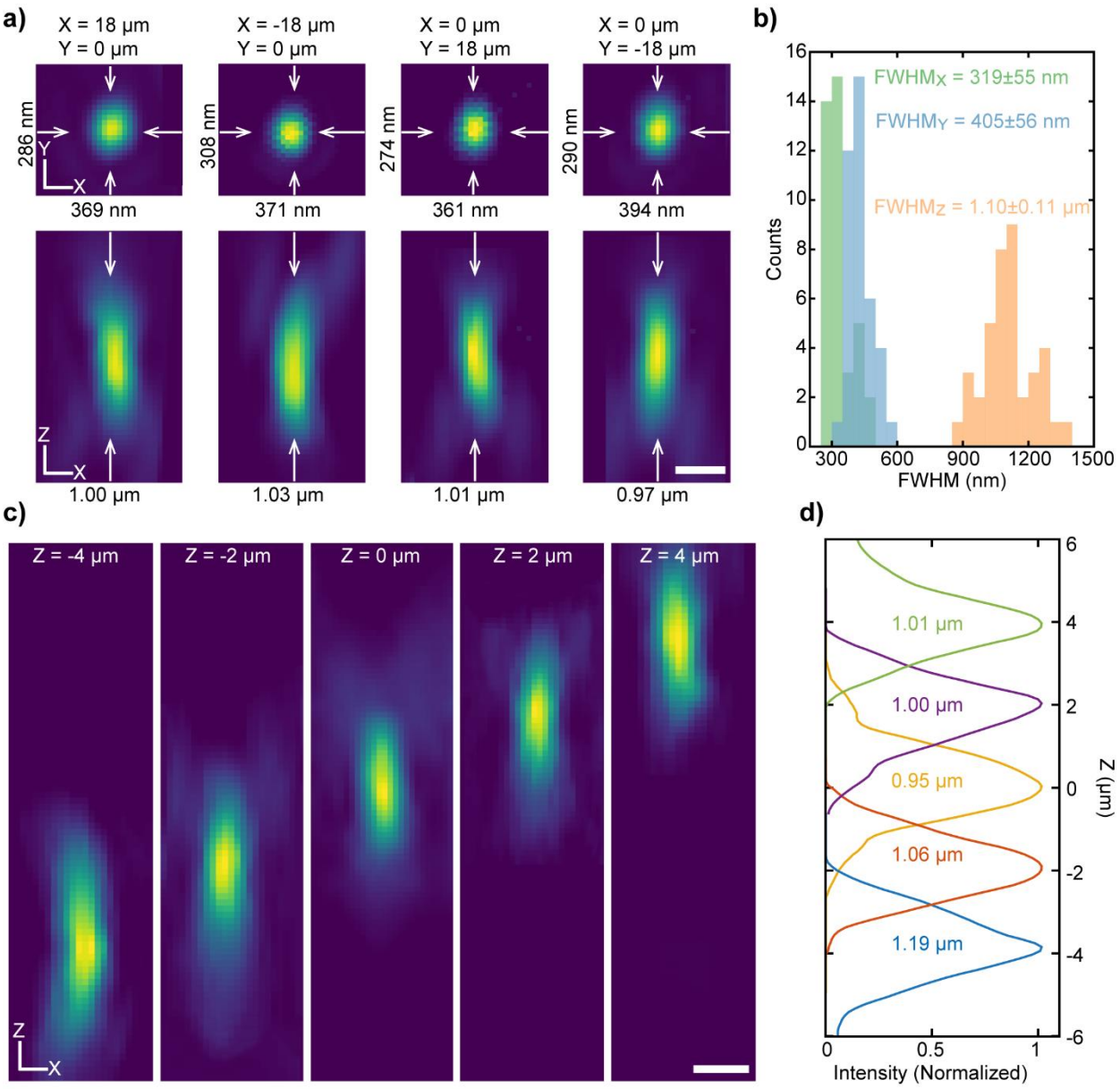


Fig. 3. 3D spatial resolution examination in various positions in 3D. (a) Images of 50 nm fluorescent green microspheres located at the positions of (18, 0), (-18, 0), (0, 18), and (0, -18) in the x-y plane. The upper panel shows the PSF in x-y plane, while the lower panel shows the PSF in x-z plane. Scale bar = 500 nm. **(b)** The statistical histogram of spatial resolution distribution in x, y, and z directions, which shows mean values of 319 ± 55 nm, 405 ± 56 nm, and 1.10 ± 0.11 μm in x, y, and z. n=39. **(c)** Fluorescent microsphere images at different z-depths (-4 μm, -2 μm, 0 μm, 2 μm, and 4 μm). Scale bar = 500 nm. **(d)** Plots of the intensity distribution of the PSFs in different axial positions in **(c).** The axial resolutions were determined by fitted FWHM.

Revision 3: 50 nm red fluorescent beads imaging result has been added to the **Supplementary Figure S6-S7:**

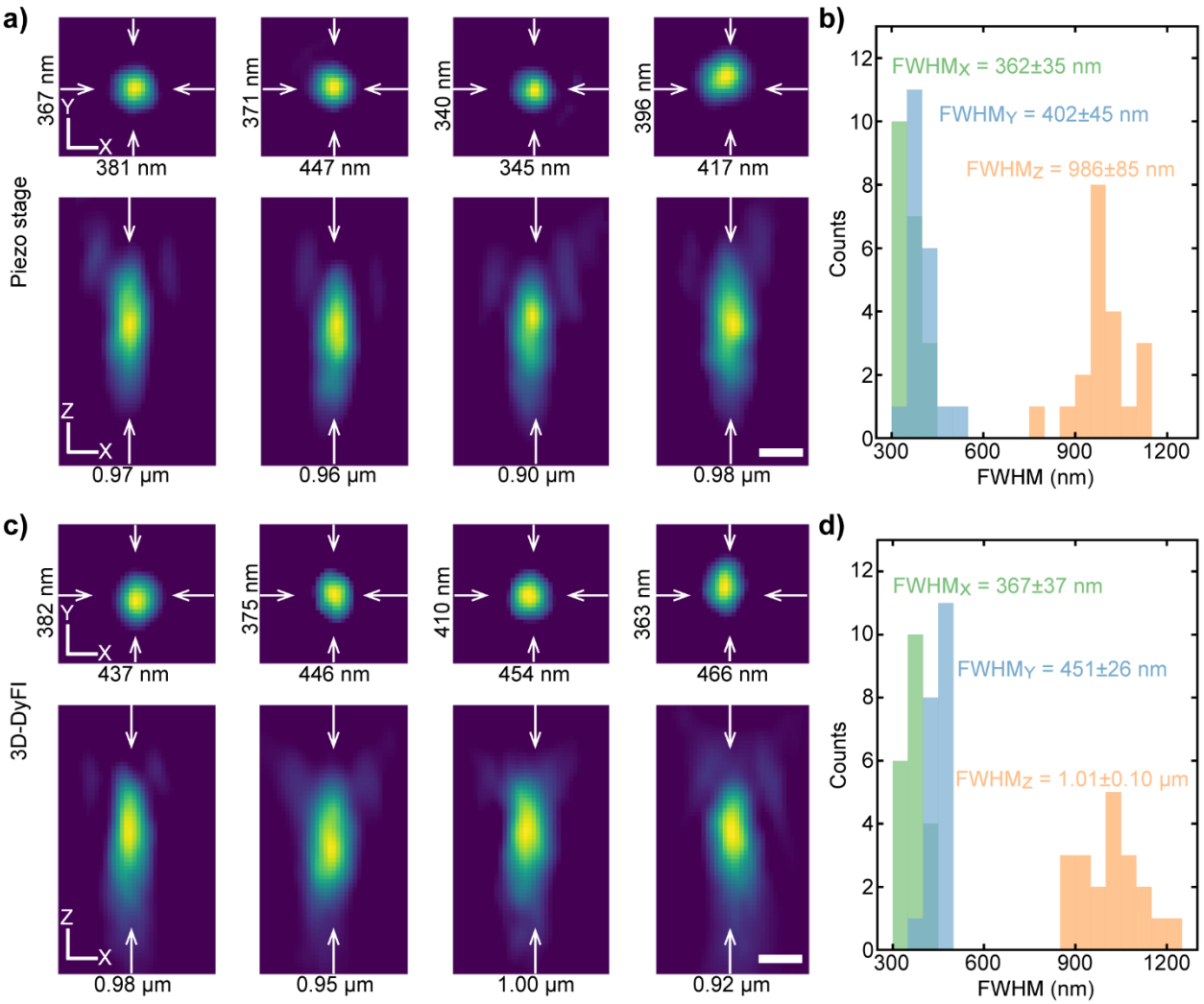


Figure S6. The resolution characterization with 50 nm red fluorescent spheres in the center of scanning range of dynamics focusing lens. Images of four typical 50 nm red fluorescent microspheres (635 nm excitation) in the x-y and x-z planes, obtained using the Piezo stage **(a-b)** and 3D-DyFI **(c-d)** systems. The upper panel shows the PSF in the x-y plane, and the lower panel shows the PSF in the x-z plane. Pixel size: 50 nm. Scale bar: 500 nm. **(b, d)** Statistical histogram of the spatial resolution distribution in the x, y, and z axis for the piezo stage-based confocal microscopy **(b)** and 3D-DyFI **(d)** systems. n = 20.

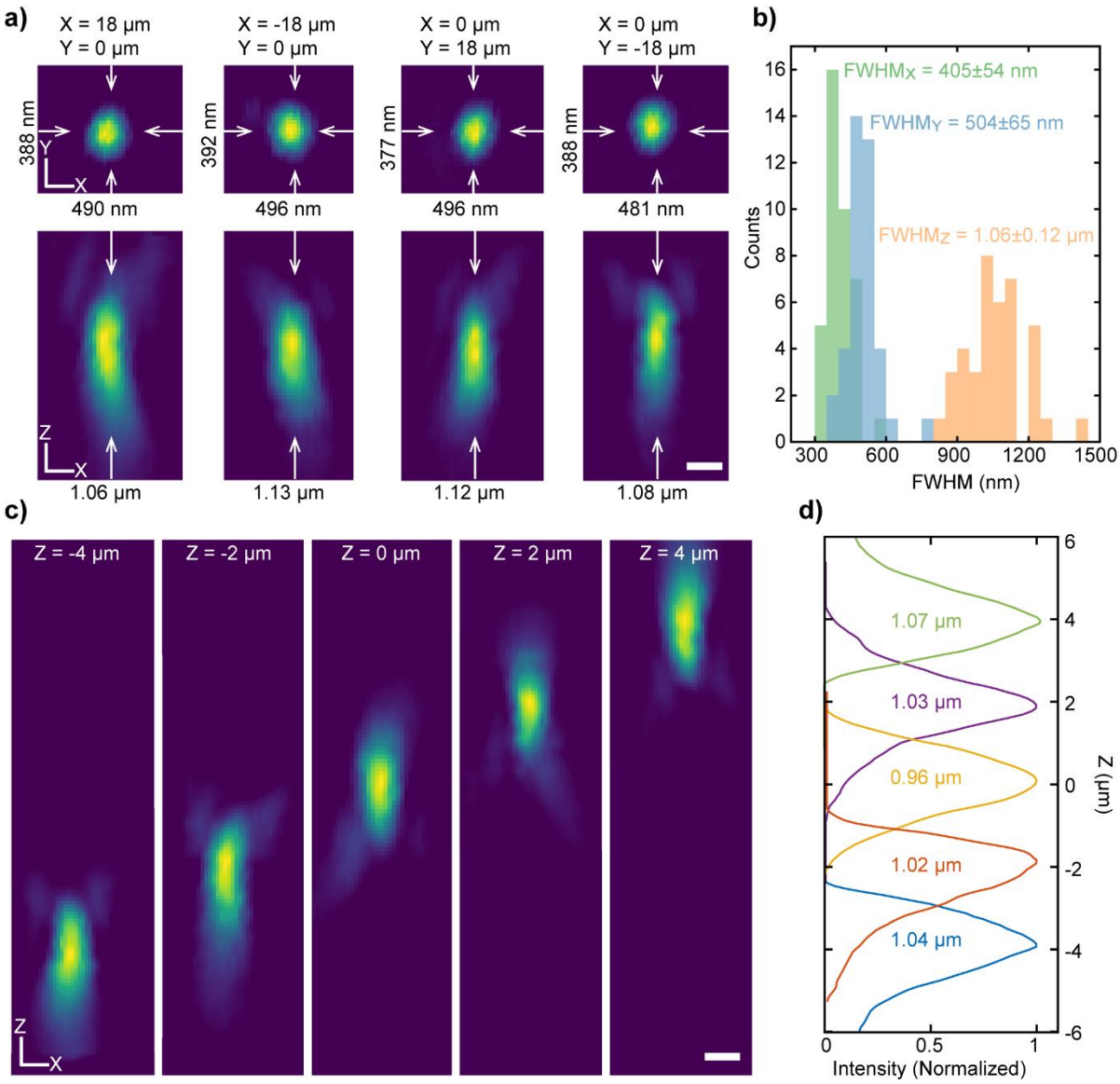


Figure S7. 3D spatial resolution examination in various positions in 3D with 50 nm red fluorescent beads. (a) Images of four typical 50 nm fluorescent microspheres located at the boundary of the maximum scanning range, with positions at (18, 0), (-18, 0), (0, 18), and (0, -18) in the x-y plane. The upper panel shows the PSF in x-y plane, while the lower panel shows the PSF in x-z plane. Scale bar = 500 nm. **(b)** The statistical histogram of spatial resolution distribution in x, y, and z directions, which shows mean values of 405 ± 54 nm, 504 ± 65 nm, and 1.06 ± 0.12 μm in x, y, and z. n=39. **(c)** Fluorescent microsphere images at different z-depths (-4 μm, -2 μm, 0 μm, 2 μm, and 4 μm). Scale bar = 500 nm. **(d)** Plot of the intensity distribution of the PSFs from **(c)** with the fitted FWHM values to determine the corresponding axial resolution.

Revision 4: The comparison of PSF profiles between 3D-DyFI and the piezo-stage based confocal microscopy also has been added to Supplementary Information:

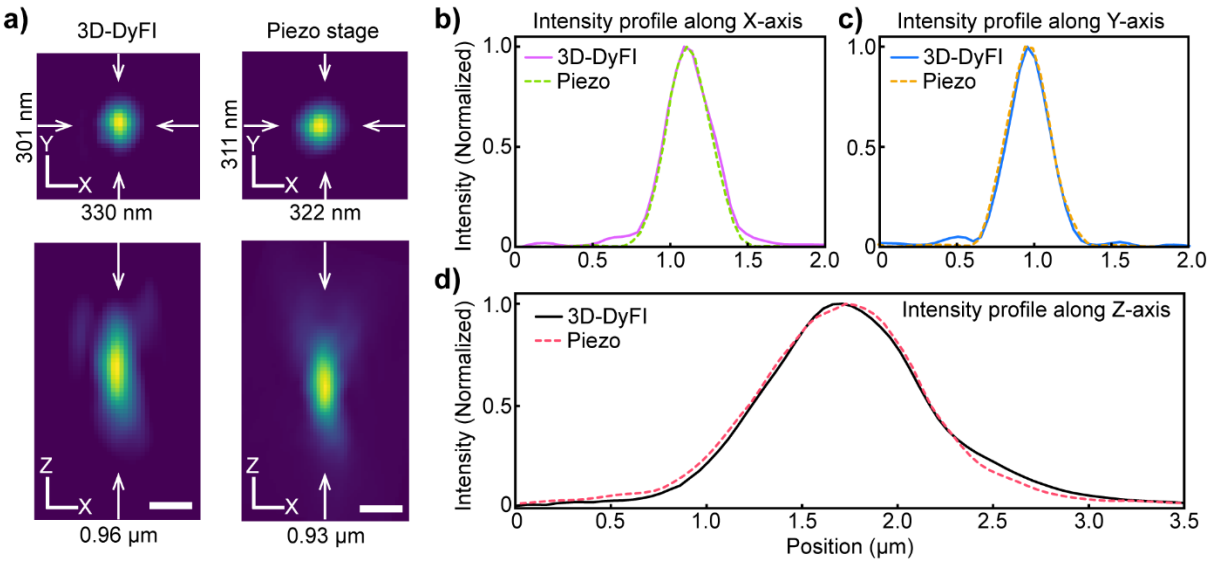


Figure S5. Comparison of the PSF between 3D-DyFI and the piezo-stage based confocal microscopy. **(a)** The scanning images of 50 nm fluorescent beads with 3D-DyFI and the piezo-stage based confocal microscopy. Upper panel: x-y plane section images; lower plane: x-z plane section images. **(b-d)** The comparative analysis of sectional intensity profiles of PSF images in **(a)** along x **(b)**, y **(c)**, and z **(d)** direction. Scale bar: 500 nm.

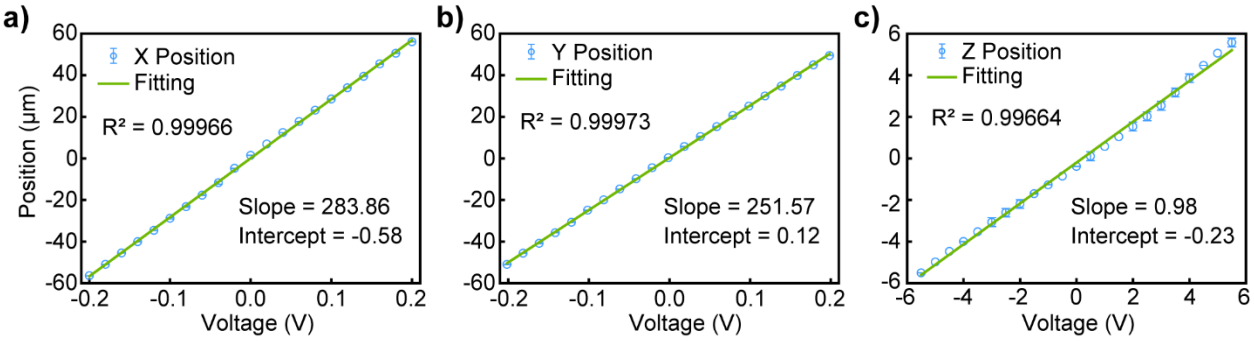
4. For the characterization of laser focus with applied voltage, the dynamic range along X, Y, Z needs to be increased. Moreover, why there are no error-bars?

Response: Thank you for your comment. We think this is a great suggestion. Upon review, we realized that incorrect data were used in the previous version of our manuscript. We sincerely apologize for this oversight.

The measurement results in previous version of our manuscript were obtained during the initial phases of assembling our experimental setup. In this context, the x and y dimensions were inaccurately measured without accounting for the bias voltage of the data acquisition card (USB-6001, National Instruments) when measuring the output voltages after 20 dB attenuators.

Additionally, the z-direction measurement was performed while the scanning tube lens was positioned differently compared to the current configuration.

In the current setup, the relationships between the applied voltages and the laser focus positions along the x, y, and z axes are shown below:



The range of applied voltage for the x and y axes is set from -0.2 V to 0.2 V. Voltages exceeding this range cause the laser spot to move beyond the aperture of the objective lens. The voltage applied to the z axis is confined to the applicable voltage range of the remote focusing system, which is from -5 V to 5 V. Although the dynamic range for the x and y axes can extend up to (-50 μm, 50 μm), significant aberrations were observed when the positions deviated substantially from the center. Consequently, we limited the operational range of x and y to (-20 μm, 20 μm) during imaging.

Each measurement was conducted three times to ensure accuracy, and error bars have been included in the plots.

Revision: The Figure 2 has been replaced with the following figure:

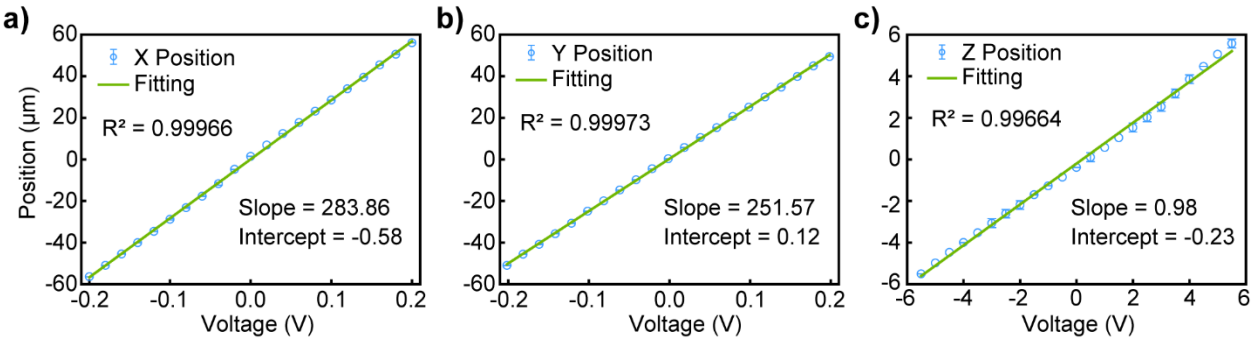


Fig. 2. The laser's focal position is linearly changed with the applied voltage. The laser focus positions as a function of applied voltage in x **(a)**, y **(b)**, and z **(c)** axes. Measured data are represented by blue circles, while the fitting results are shown with green lines. n = 3.

5. What is the minimum sampling for X, Y, Z positions that can be achieved.

Response: Thank you for your comment. To measure the minimum sampling of 3D-DyFI, we directed the laser to a target plate over a long distance. By varying the voltage applied to the dynamic focusing system in the x and y directions and monitoring the resultant spot motion, we determined that the minimum responsive voltage is 0.0705 mV, which corresponds to a displacement of 20 nm at the imaging focal plane. It should be noted that a 20 dB attenuator is positioned between the output of the FPGA data acquisition card and the input to the dynamic focusing system. The resolution of data acquisition card (PCIe-7858, NI) analog output is 16 bit with a ± 10 V output range. Consequently, its analog voltage output resolution is 0.3 mV, which can be enhanced to 0.003 mV using a 20 dB attenuator.

No attenuator was used for z-axis control. While the responsive voltage can reach 0.0705 mV, the resolution is constrained by the analog voltage output resolution of the FPGA data acquisition card, which is 0.3 mV. Referring to Figure 2, the minimum sampling distance in the z-axis is determined to be 0.29 nm.

In summary, the minimum sampling distance is 20 nm in the x and y directions and 0.29 nm in the z direction.

6. The claim, "… uniform resolution across a decent 3D scanning (40μm x 40μm x 10μm). " needs further ***discussion****.*

Response: Thank you for your comment. We acknowledge that our previous description of "uniform resolution across a decent 3D scanning volume" is inaccurate. The best resolution is achieved in the center of scanning range of 3D dynamic focusing system, while deviating from the center resulting in resolution degradation. This effect is more obvious in axial direction (from 0.95 μm to 1.19 μm). It could be modified to "relatively uniform". For optimal 3D imaging using

3D-DyFI, it is advisable to position the imaging target near the center of the scanning range of the 3D dynamic focusing system.

Revision: We have revised the “uniform” to “relatively uniform” and a discussion has been added:

“These results demonstrate that 3D-DyFI maintains relatively uniform resolution across a decent 3D scanning volume (40 μm × 40 μm × 10 μm). It is important to note that optimal resolution is achieved at the center of the scanning range of the 3D dynamic focusing system. Deviations from the center lead to resolution degradation, an effect that is more pronounced in the axial direction. Consequently, for optimal 3D imaging using 3D-DyFI, it is advisable to position the imaging target near the center of the scanning range of the 3D dynamic focusing system.”

7. In general, the calibration and testing part is OK but the application part is weak. This can be addressed by adding imaging additional cell samples and incorporating the same in the revised version.

Response: Thank you for your nice suggestion. We have extended the application of 3D-DyFI to encompass a broader range of cellular imaging, including mitochondria, lipid droplets, and microtubules. The imaging results have been included in **Figure 6**, **Supplementary Figure 8-10**.

Revision: The following figures have been added:

To further demonstrate the 3D cellular imaging capability of 3D-DyFI. We imaged mitochondria (**Fig. 6**), cell microtubule (**Supplementary Fig. S8**), cell membrane (**Supplementary Fig. S9**) and lipid droplets in live cell (**Supplementary Fig. S10**). These results demonstrate that the 3D-DyFI can be employed for 3D volume cellular imaging.

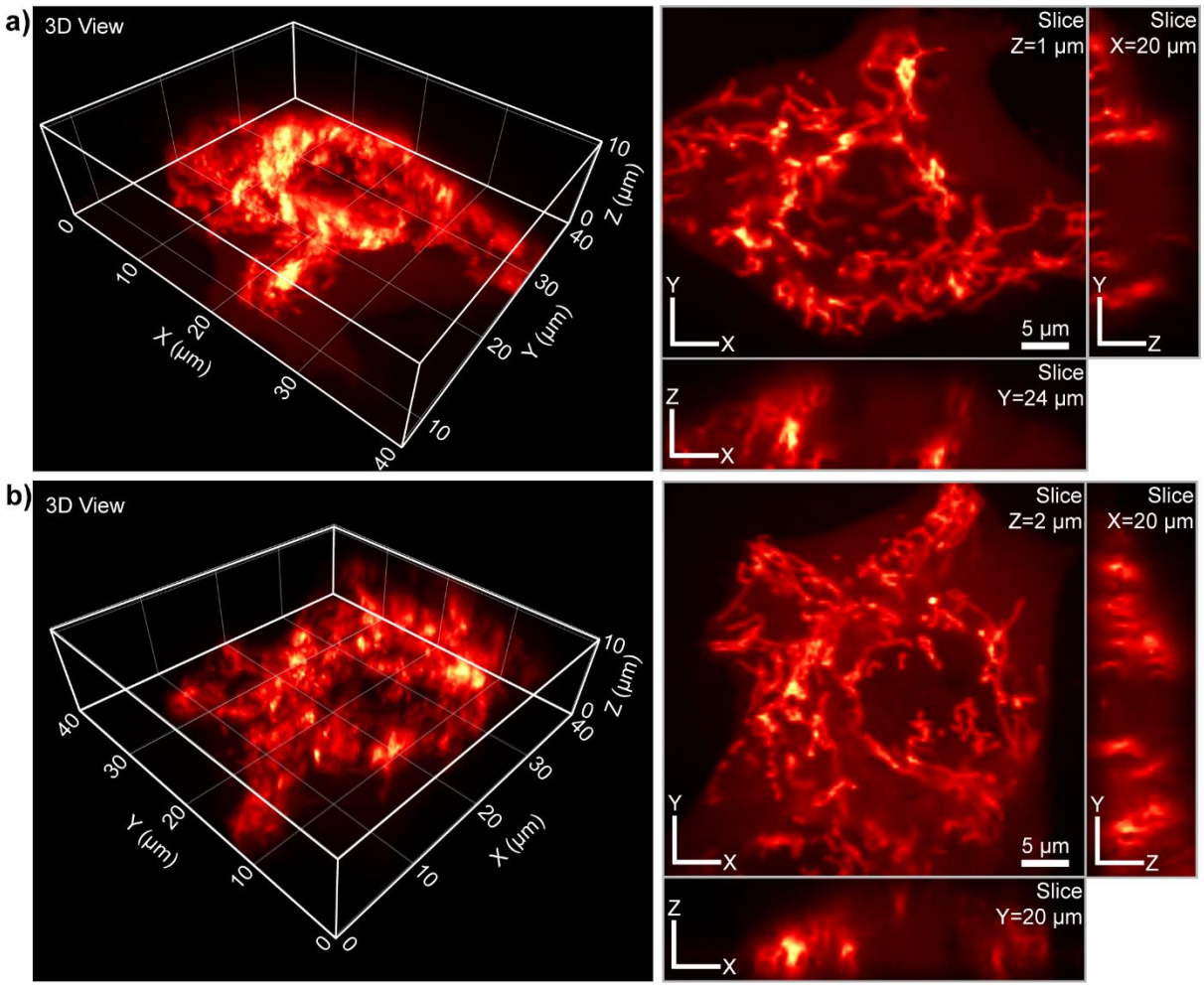


Fig. 6. 3D imaging of mitochondria of live Hela cells with 3D-DyFI. (**a-b**) The mitochondria were labeled with mGold fluorescent protein. The left panel shows 3D live cell imaging. The right panel shows the section image in the x-y, x-z, and y-z planes. The sectioning positions are labeled in the upper right corner of the images.

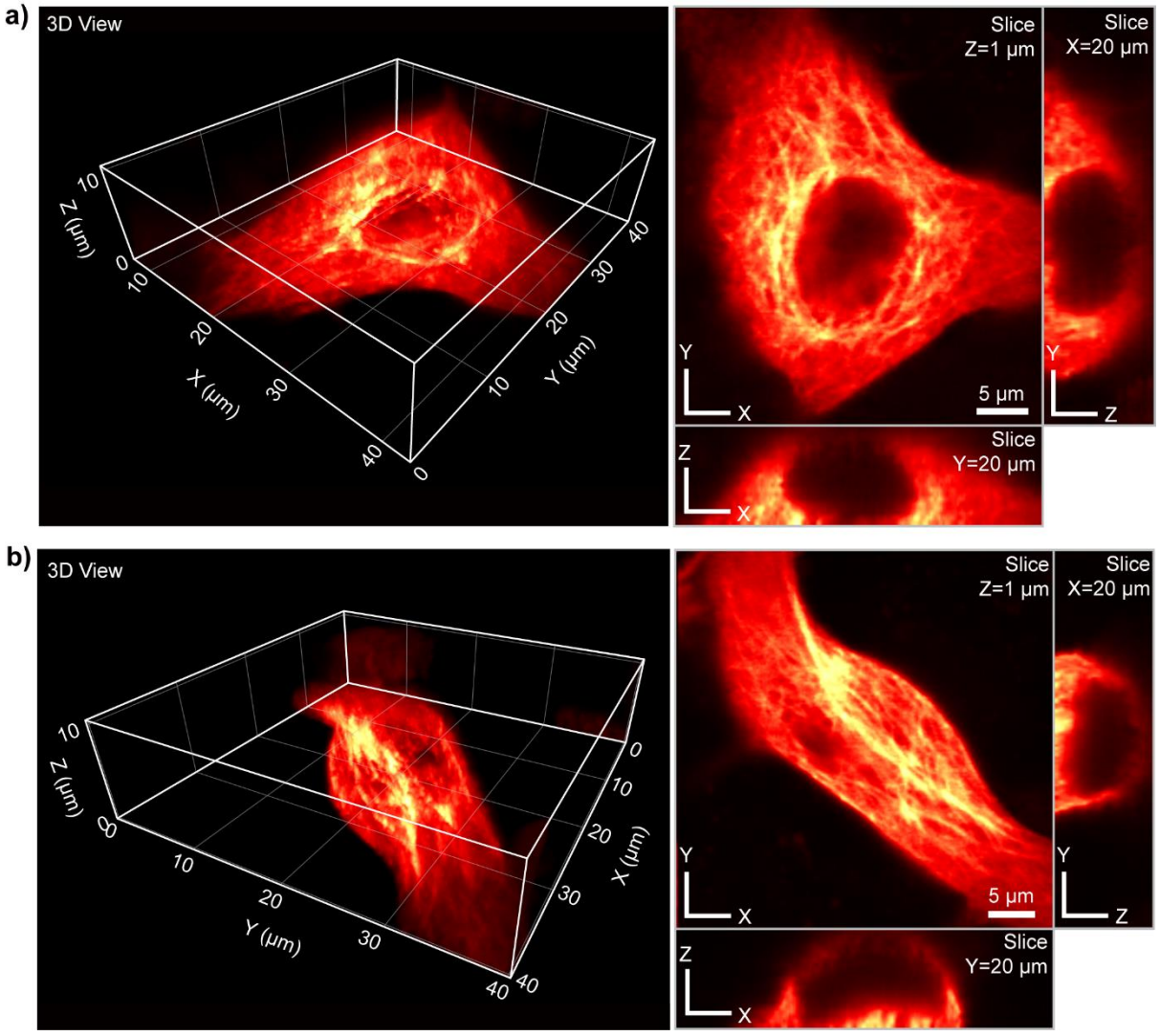


Figure S8. 3D imaging of microtubules of live Hela cell. The microtubules were labeled with mBaojin fluorescent protein. The left panel shows 3D live cell imaging. The right panel shows the section image in the x-y, x-z, and y-z planes. The sectioning positions are labeled in the upper right corner of the images.

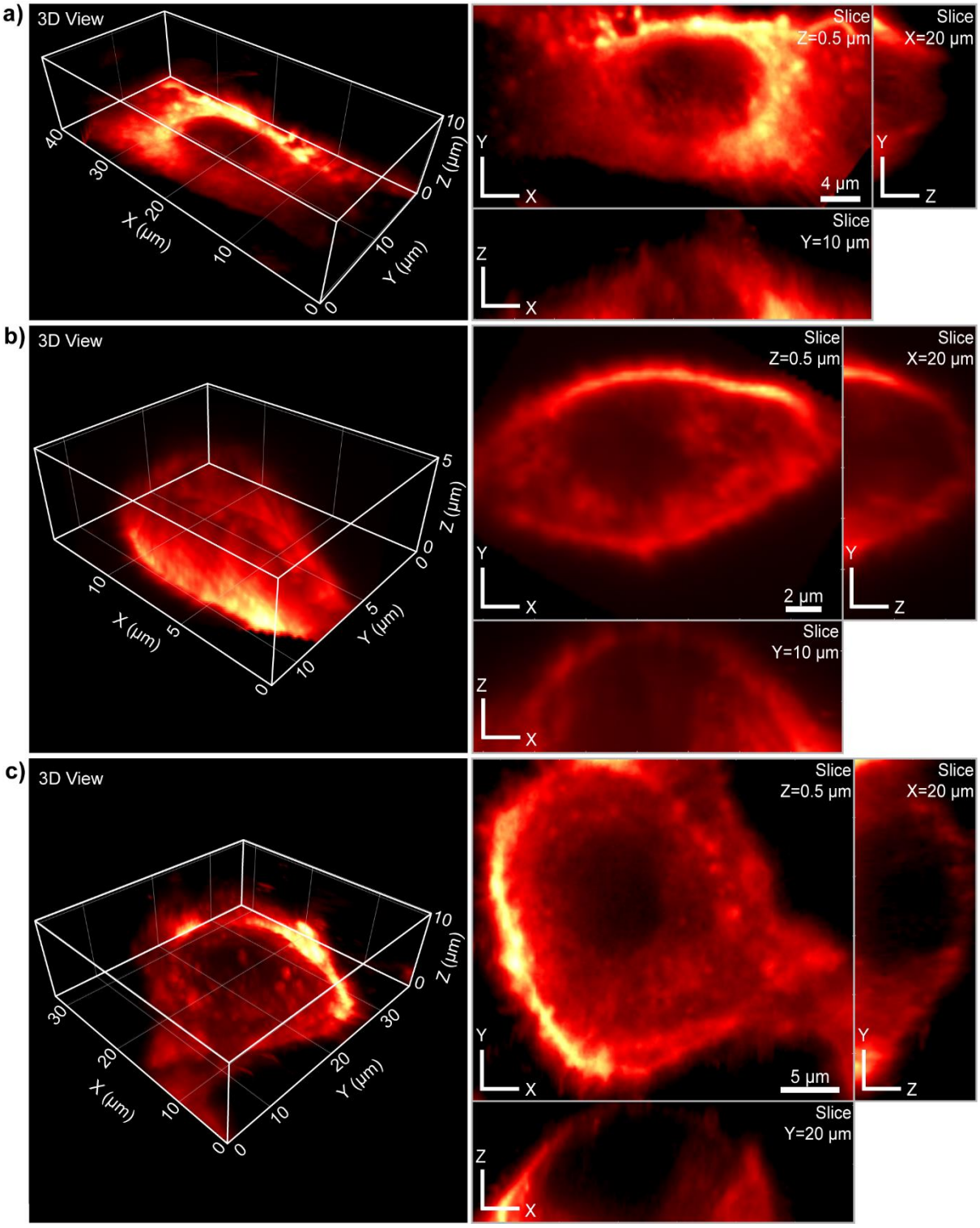


Figure S9. Three-dimensional imaging of DiD-stained Hela cell membranes with 3D-DyFI (a-c). The left panel shows 3D live cell imaging. The right panel shows the section image in the x-y, x-z, and y-z planes. The sectioning positions are labeled in the upper right corner of the images.

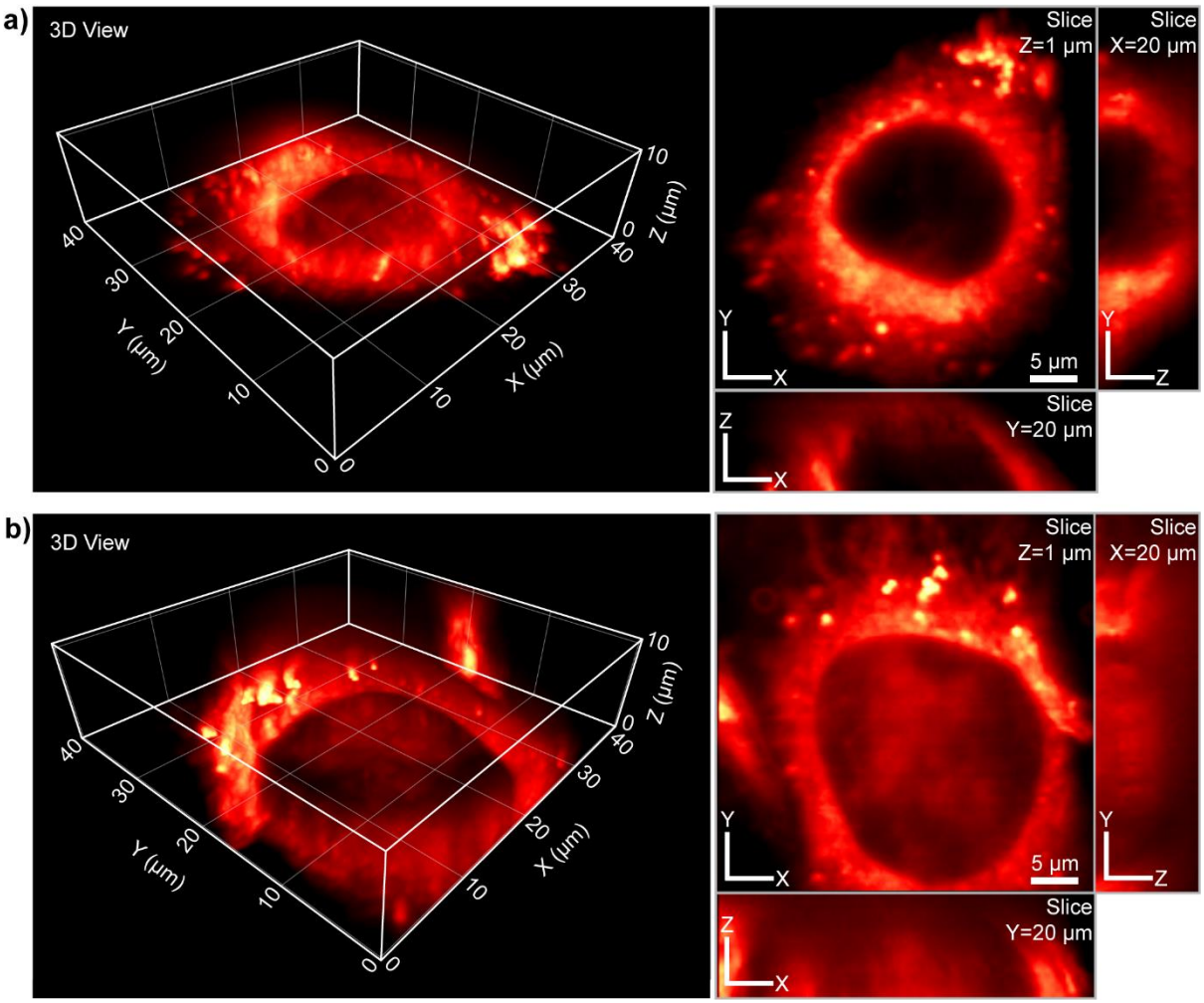


Figure S10. 3D imaging of Nile Red labeled live L929 cell. The lipid droplets (bright spots) and cell membrane were labeled by Nile Red. The left panel shows 3D live cell imaging. The right panel shows the section image in the x-y, x-z, and y-z planes. The sectioning positions are labeled in the upper right corner of the images.

8. The images shown in Fig. 5 appears to be blurred. Was the system out of focus while taking data? Certainly, this needs explanation.

Response: Thank you for your comment. This is a good question. We believe the observed blurring in this figure is attributable to the use of large pixel sizes. Specifically, in the piezo stage-based confocal imaging presented in this figure, the pixel sizes were 1 μm in both the x and y dimensions, and 0.5 μm in the z dimension. For 3D-DyFI, the pixel size in x, y, and is 1 μm. We acknowledge that these pixel sizes were excessively large and inconsistent between the

two systems. We apologize for this oversight. The **Fig. 5** has since been updated with newly acquired data with smaller pixel sizes in the current version of the manuscript to address this issue.

Revision: This **Fig. 5** has been updated:

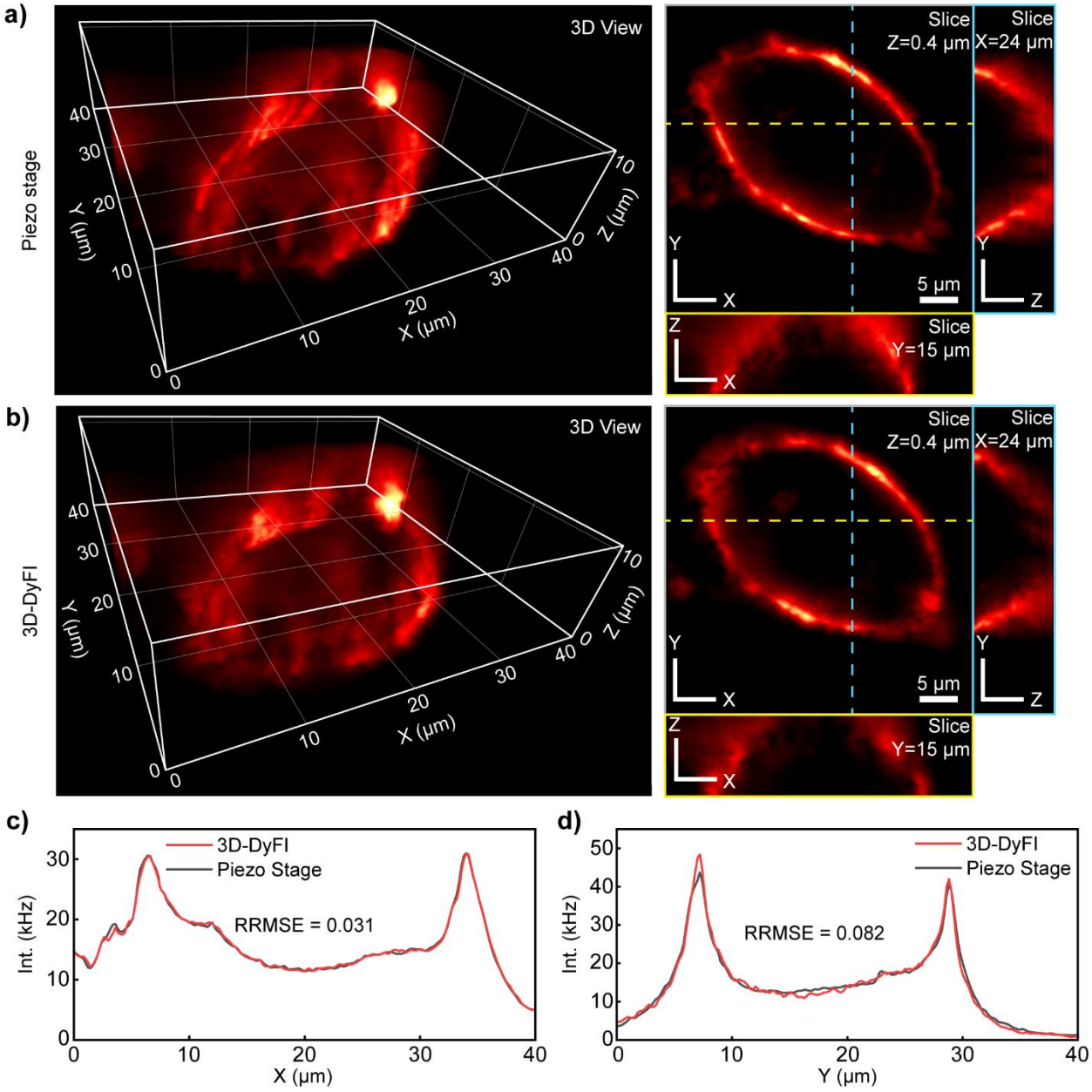


Fig. 5. Three-dimensional imaging of DiD-stained Hela cell membranes with 3D-DyFI and piezo stage-based confocal microscopy. (a) 3D live cell imaging with piezo stage-based

confocal microscopy. **(b)** 3D live cell imaging with 3D-DyFI. The corresponding section images in the x-y, x-z, and y-z planes are shown in the right panel. The x-y plane section images were taken at the axial position of z = 0.4 μm; the x-z plane section images were taken at the position of y = 15 μm; and the y-z plane section images were taken at the position of x = 24 μm. Dashed lines at x = 24 μm (cyan) and y = 15 μm (yellow) are marked on the x-y plane image. **(c-d)** The intensity profiles along the y = 15 μm **(c)** and x = 24 μm lines **(d)**, respectively. The Relative root mean squared errors (RRMSE) between the two intensity profiles were calculated, resulting in 0.031 and 0.082, respectively. Pixel size in x and y: 0.2 μm; Pixel size in z: 0.4 μm.

9. The authors state that, the illumination time was increased to 5 ms and 10 ms. Show 3D images to visually access image quality.

Response: Thank you for your suggestion. We have performed the 3D imaging of 50 nm red fluorescent microspheres with 2 ms. 5 ms, and 10 ms illumination time. The results have been added to the supplementary information.

Revision: The 3D images of 50 nm red fluorescent microspheres at different illumination times have been added to the **Supplementary Figure S12**:

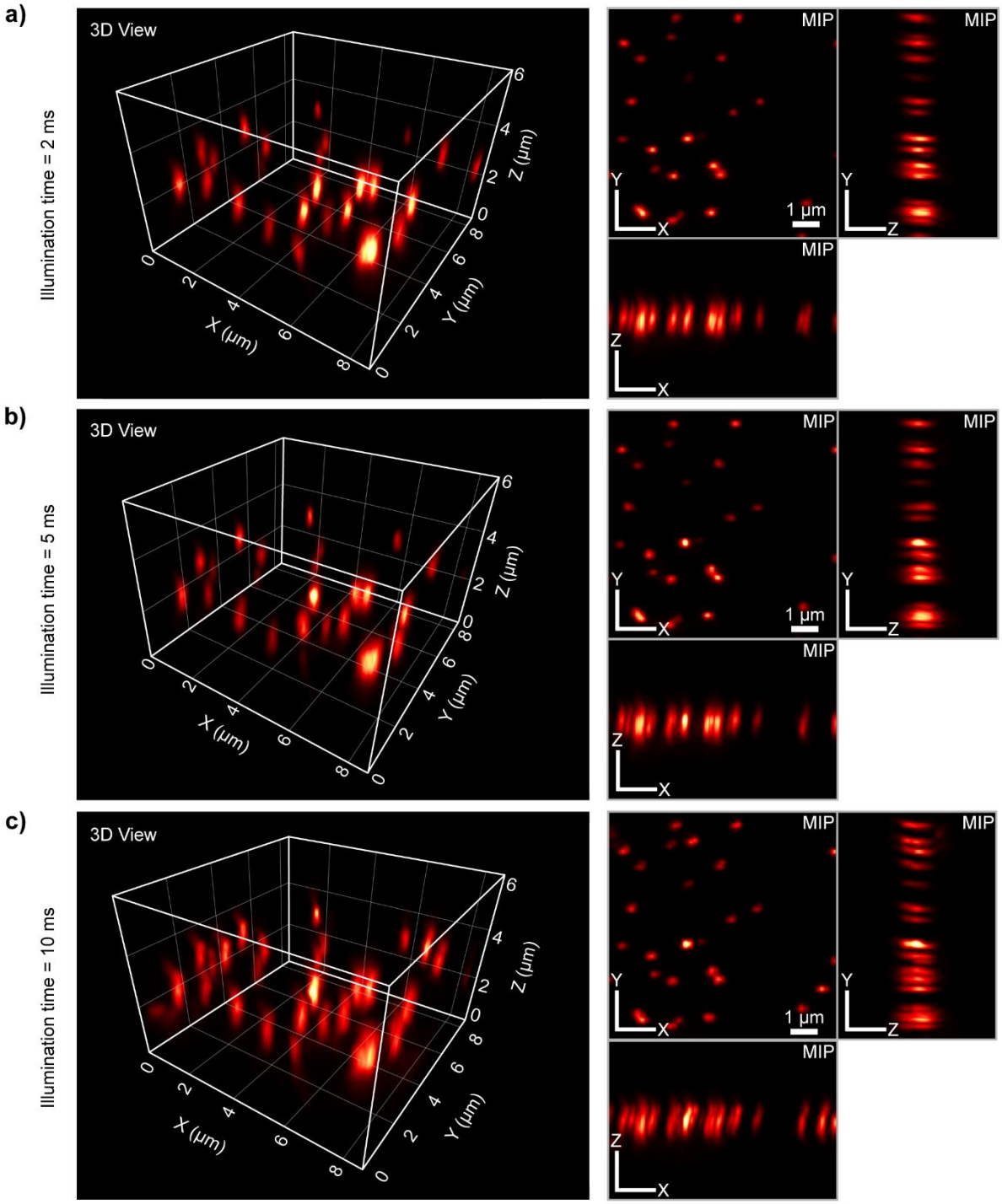


Figure S12. 3D imaging of 50 nm red fluorescent microspheres with different pixel illumination times. Microspheres located at coverslip were imaged with 3D-DyFI with illumination times of 2 ms **(a)**, 5 ms **(b)**, and 10 ms **(c)**. The left panels show 3D views, while the right panels display maximum intensity projection (MIP) images along the x-y, y-z, and x-z planes. Scale bar: 1 μm.

10. In conclusion section, the claim "… while maintaining excellent spatial resolution …" need to be substantiated with additional metric / sectional intensity plots.

Response: Thank you for your comments. We think the word of "excellent" may not be appropriate here. We have modified this sentence to "…while maintaining the spatial resolution comparable to piezo stage-based confocal microscopy…". In Supplementary Figure S5, we have included the comparison of sectional intensity profiles between 3D-DyFI and the piezo-stage based confocal microscopy. The results indicate that 3D-DyFI achieves spatial resolution comparable to that of the piezo-stage based confocal microscopy.

Revision: The following figure has been added to the Supplementary Information:

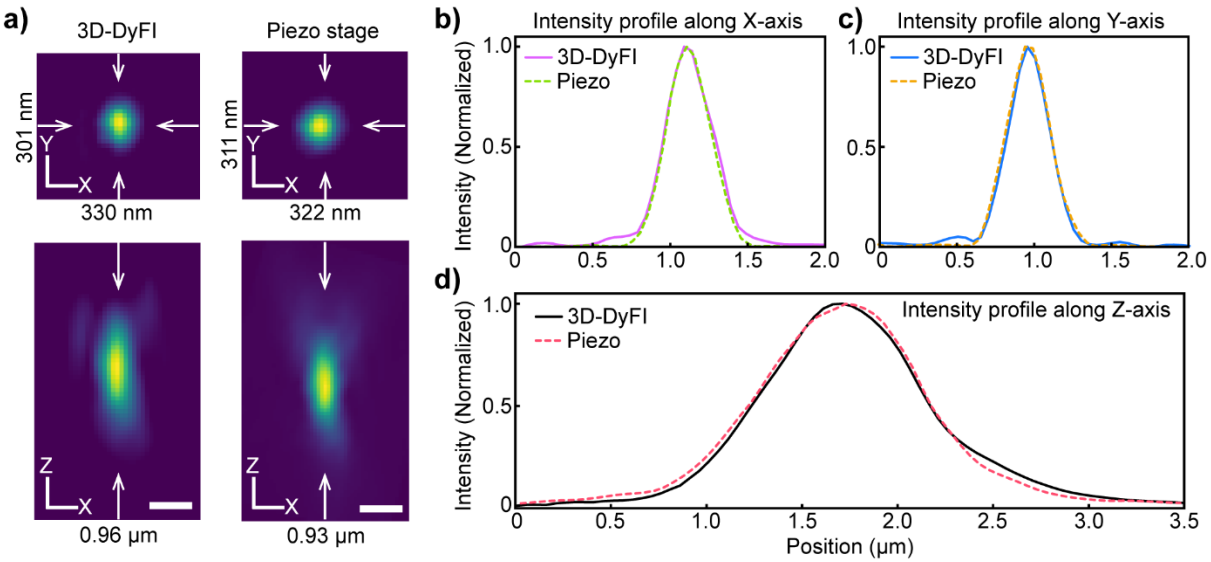


Figure S5. Comparison of the PSF between 3D-DyFI and the piezo-stage based confocal microscopy. **(a)** The scanning images of 50 nm fluorescent beads with 3D-DyFI and the piezo-stage based confocal microscopy. Upper panel: x-y plane section images; lower plane: x-z plane section images. **(b-d)** The comparative analysis of sectional intensity profiles of PSF images in **(a)** along x **(b)**, y **(c)**, and z **(d)** direction. Scale bar: 500 nm.

Minor:

1. In the introduction section, authors state large imaging depth of CSLM. This is not correct. It is the Bessel-beam based microscopy techniques that are capable of large imaging depths.

Response: Thank you for your comments. We agree with you that Bessel-beam based microscopy techniques, such as Bessel-beam based light-sheet microscopy, are more suitable for imaging over large depths. In here we wanted to express that the CSLM has relatively larger imaging depth compared with conventional epifluorescence microscopy and TIRF microscopy. To avoid any ambiguity, we have removed the "large imaging depth" in manuscript.

Revision: The following words have been removed in the introduction section:

"and large imaging depth".

2. Details of filters, APD and other optical components are missing.

Response: Thank you for your comments. We apologize for our oversight. The details of these components have been described in the revised manuscript.

Revision: The details of the filters, dichroic mirrors, and avalanche photodiode (APD) have been incorporated into the section on Configuration and 3D Resolution Characterization of 3D-Dynamic Focusing Imaging (3D-DyFI). Additionally, the specifics of the dynamic focusing system are described in Supplementary Figure 1 and Supplementary Note 1.

3. In general, the language needs improvement. For example: "… then the sample was then moved…".

Response: Thank you for your comments. We have meticulously refined the language in the revised manuscript to ensure clarity and precision.

4. Few sentences need to be rewritten and with more clarity. For example: " … imaging temporal resolution …"; "… and then moved laser focus 1 μm away…", "In previously frequently reported 3D …" etc.

Response: Thank you for your comments. We have revised the descriptions in the updated manuscript accordingly. We apologize once again for the deficiencies in the writing of the previous version of the manuscript.

Overall, the work has potential, but more needs to be done to bring out the uniqueness of the proposed technique. In view of this, the manuscript may be considered for publication provided the comments are addressed and additional data is provided.

Response: We sincerely appreciate your valuable feedback.

We sincerely thank the editor and all reviewers for their valuable feedback that we have used to improve the quality of our manuscript. The reviewers' comments are laid out below in *italics* and specific concerns have been numbered. Our response is given in blue font and changes or additions to the manuscript are given in green font. The corresponding revisions are highlighted in the revised manuscript.

Reviewer #1 (Remarks to the Author):

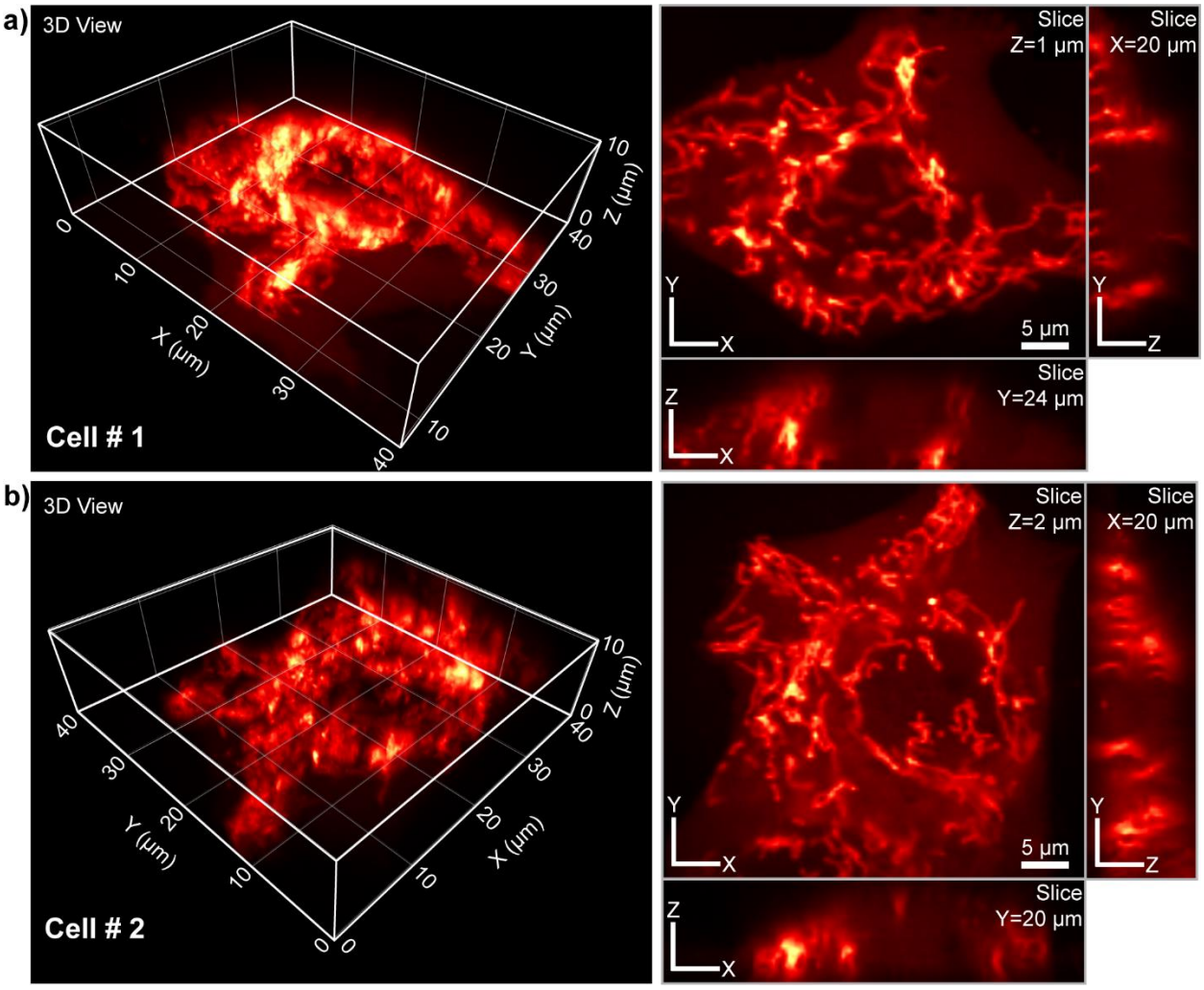
The authors have addressed the important concerns raised in the original manuscript and significantly improved the manuscript. I recommend its publication for Communications Engineering.

Response: Thank you very much for your positive comments and recommendation. We greatly appreciate your professional review of our manuscript.

One minor comment: Please revise the legend in Fig. 6 to clarify how different (a) and (b) are.

Response: Thank you for your comment. We have revised the caption and legend in the revised Figure 5 (formerly Figure 6) to explicitly state that panels (a) and (b) present images from two different cells.

Revision: Fig. 5 and its caption have been revised:



“**Fig. 5. 3D imaging of mitochondria of live HeLa cells with 3D-DyFI.** The mitochondria were labeled with mGold fluorescent protein. The left panel shows 3D live cell imaging. The right panel shows the section image in the x-y, x-z, and y-z planes. The sectioning positions are labeled in the upper right corner of the images. (a) and (b) are obtained from two different cells”

Reviewer #2 (Remarks to the Author):

The authors addressed my questions and I think that the manuscript is ready for publication in the current form.

Response: Thank you very much for your positive comments and recommendation. We greatly appreciate your professional review of our manuscript.

Reviewer #3 (Remarks to the Author):

The authors have addressed my queries adequately. I recommend the paper for publication pending the following small query.

Response: Thank you very much for your positive comments and recommendation. We greatly appreciate your professional review of our manuscript.

1. Figure 2 in the main manuscript may be either merged with figure 1 or shifted to Supplementary materials.

Response: Thank you for your kind suggestion. Now we have merged Fig. 1 and Fig. 2 into a new Fig. 1 in the revised manuscript.

Revision: Fig. 1 and Fig. 2 have been consolidated into a new Fig. 1 in the revised manuscript:

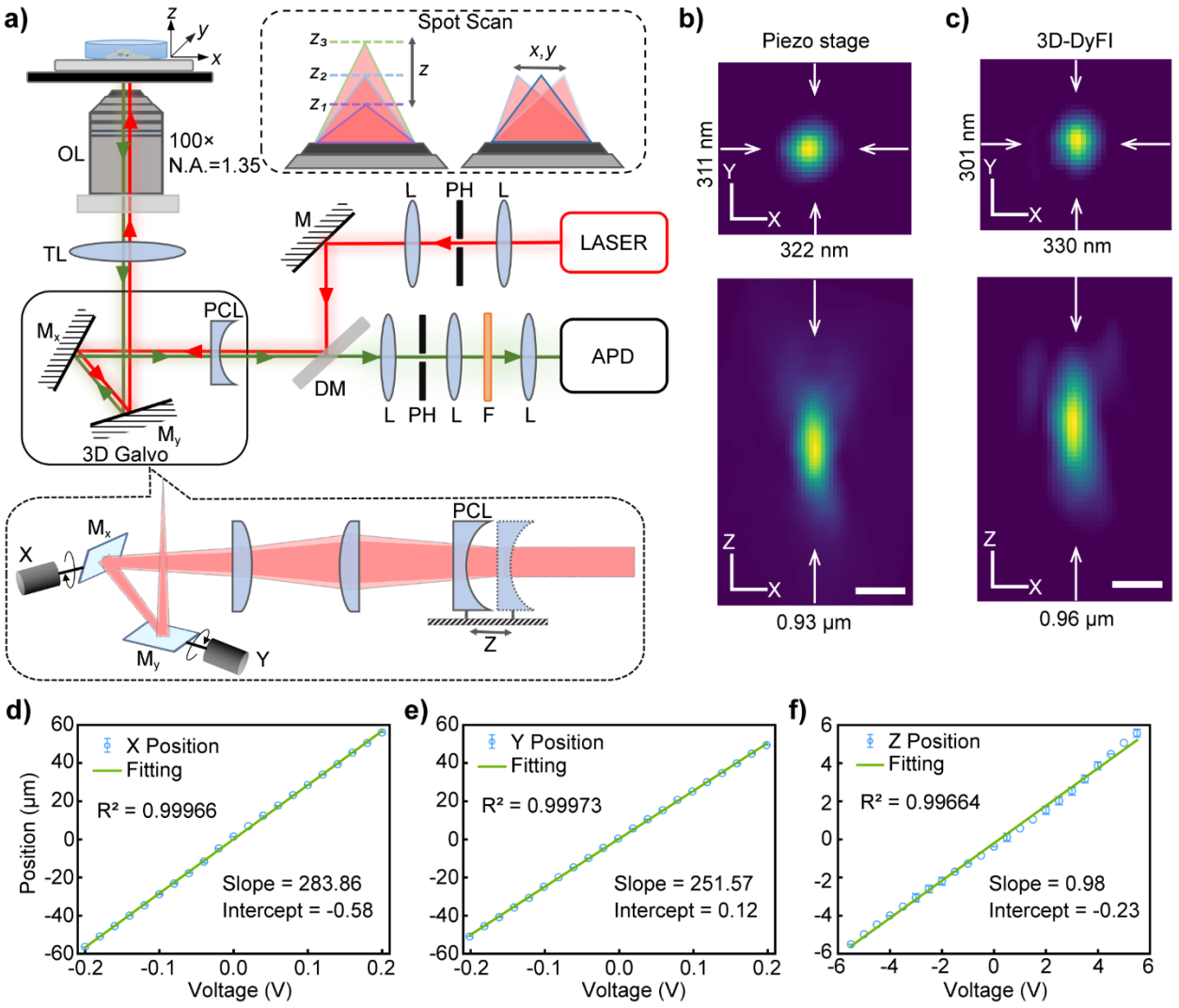


Fig. 1. Schematic, point spread function, and the linearity characterization of the 3D-DyFI system. (a) Schematic of 3D-DyFI. OL: Objective Lens, M: Mirror, F: fluorescence filter, DM: Dichroic Mirror, PH: Pinhole, PCL: Plano-Concave Lens, APD: Avalanche Photodiode. **(b-c)** The x-y scanning images (up panel) and x-z scanning images (low panel) of 50 nm green fluorescent microsphere with **(b)** piezoelectric stage scanning and **(c)** 3D-DyFI. Scale bar: 500 nm. **(d-f)** The laser focus positions as a function of applied voltage in x **(d)**, y **(e)**, and z **(f)** axes. Measured data are represented by blue circles, while the fitting results are shown with green lines. n = 3.

2. In the figure on page #28 of the rebuttal letter (which is Fig. 2C in the main manuscript), the authors show the plots for which the horizontal axis is in Volt. Is this true. Should it be in milli-Volts. Please check. This gives an erroneous minimum sampling of 0.29 nm i.e, 2.9 Angstrom.

Also note that the voltage output resolution of FPGA card is 0.3 mV. How the plots are shown in Volts.

Response: Thank you for your comment. The unit on the horizontal axis in Fig. 2C is indeed volts, with the operational voltage range approximately ±5 V, corresponding to an axial position change of about ±5 μm as illustrated in Fig. 2C. Given the voltage output resolution of the FPGA card is 0.3 mV and the slope is 0.98 in Fig. 2C, the minimum sampling distance along the z-axis is calculated to be 0.29 nm. Experimentally measuring a 0.29 nm sampling distance is challenging. In practical applications such as 3D random-access imaging or 3D volumetric imaging, the minimum axial step size used in this study is 100 nm. Nonetheless, the capacity for such fine minimum sampling on the z-axis may offer advantages for other types of studies, such as interference measurement.

Review Report:

The manuscript titled, “three dimensional random-access confocal microscopy with 3D remote focusing system” by Li et al discuss a rapid 3D random access remote focusing technique for volume imaging. The technique claims multifold improvement in response time along x, y, z axes. Moreover, the technique achieves scanning without physically moving the sample and without mechanical movement. This is good for a range of applications. While the concept is good and economical, the application part needs to improve for the advantages of the technique to be apparent. Here are my major and minor comments:

Major:

1. It would be nice to see the depth imaging capability of the proposed system on a cellular system.
2. The schematic diagram needs more discussion with special attention to the 3D galvo section. Specifically discuss how PCL on linear actuator and Galvo mirror operated synchronously during data acquisition.
3. Needs more detail on the resolution of the proposed system with additional data.
4. For the characterization of laser focus with applied voltage, the dynamic range along X, Y, Z needs to be increased. Moreover, why there are no error-bars ?
5. What is the minimum sampling for X, Y, Z positions that can be achieved.
6. The claim, “... uniform resolution across a decent 3D scanning (40μm x 40μm x 10μm). ” needs further discussion.
7. In general, the calibration and testing part is OK but the application part is weak. This can be addressed by adding imaging additional cell samples and incorporating the same in the revised version.
8. The images shown in Fig. 5 appears to be blurred. Was the system out of focus while taking data ? Certainly, this needs explanation.
9. The authors state that, the illumination time was increased to 5 ms and 10 ms. Show 3D images to visually access image quality.

10. In conclusion section, the claim "… while maintaining excellent spatial resolution …" need to be substantiated with additional metric / sectional intensity plots.

Minor:

1. In the introduction section, authors state large imaging depth of CSLM. This is not correct. It is the Bessel-beam based microscopy techniques that are capable of large imaging depths.
2. Details of filters, APD and other optical components are missing.
3. In general, the language needs improvement. For example: "… then the sample was then moved…".
4. Few sentences need to be rewritten and with more clarity. For example: " … imaging temporal resolution …"; "… and then moved laser focus 1 μm away…", "In previously frequently reported 3D …" etc.

Overall, the work has potential, but more needs to be done to bring out the uniqueness of the proposed technique. In view of this, the manuscript may be considered for publication provided the comments are addressed and additional data is provided.